resource_util vs resource_util_max

Side-by-Side Comparison

COMP 597 — Energy Profiling of Whisper Fine-Tuning

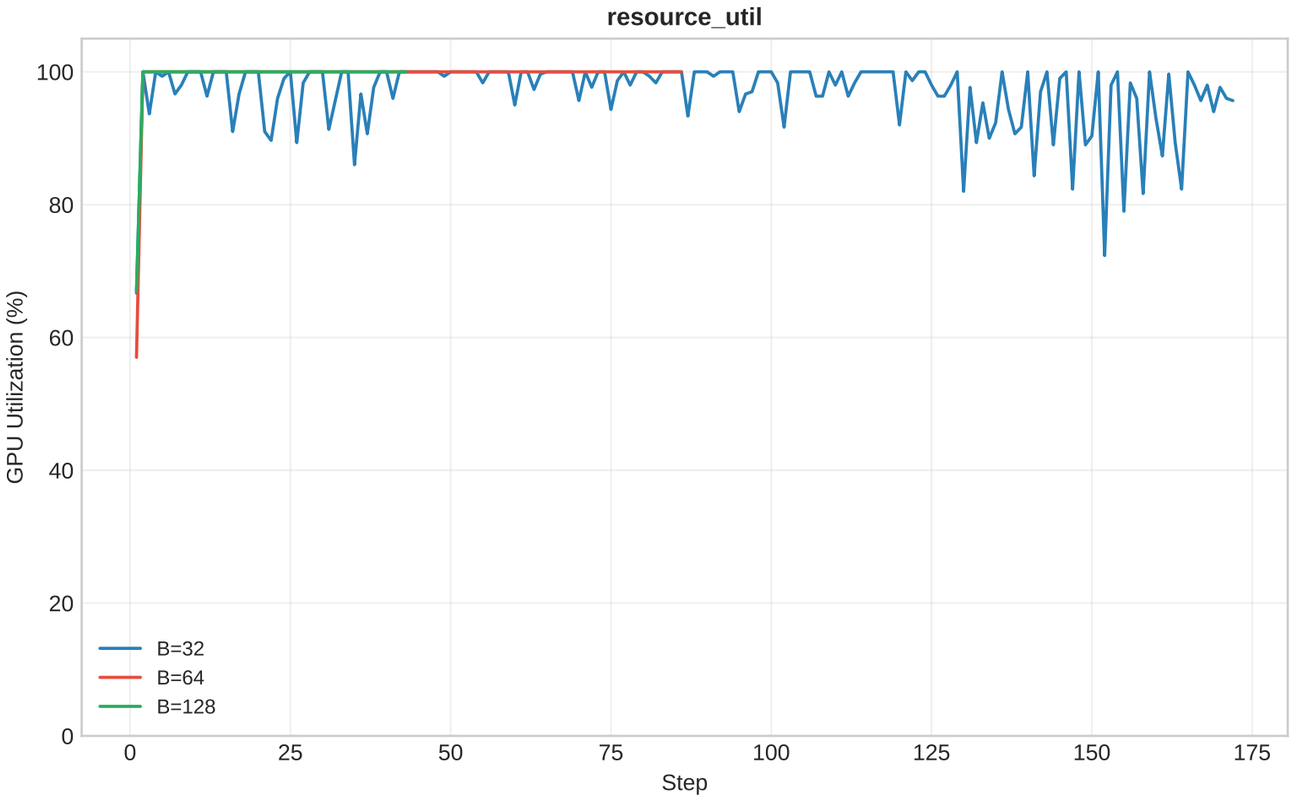
resource_util: per-step GPU/CPU metrics with CUDA synchronization
resource_util_max: per-step GPU/CPU metrics without CUDA sync (async max readings)

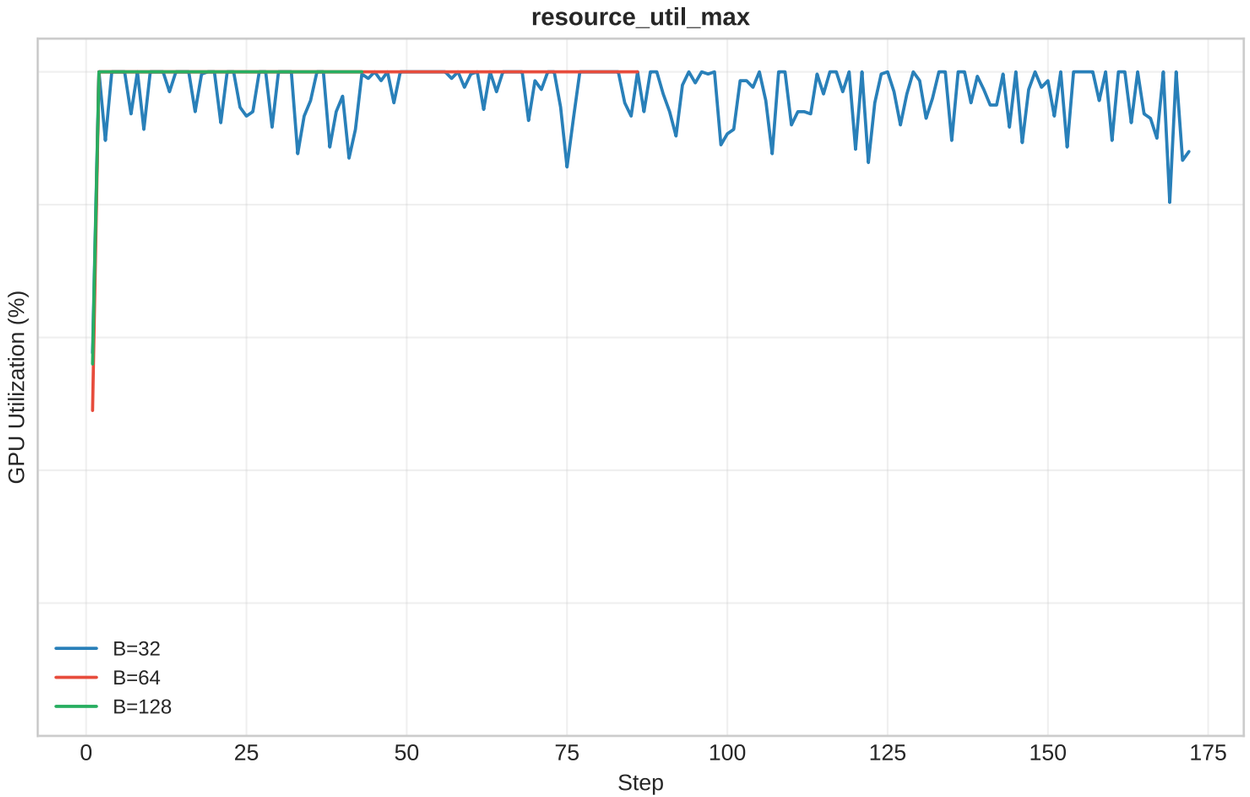
Green = resource_util | Blue = resource_util_max

Summary: resource_util vs resource_util_max

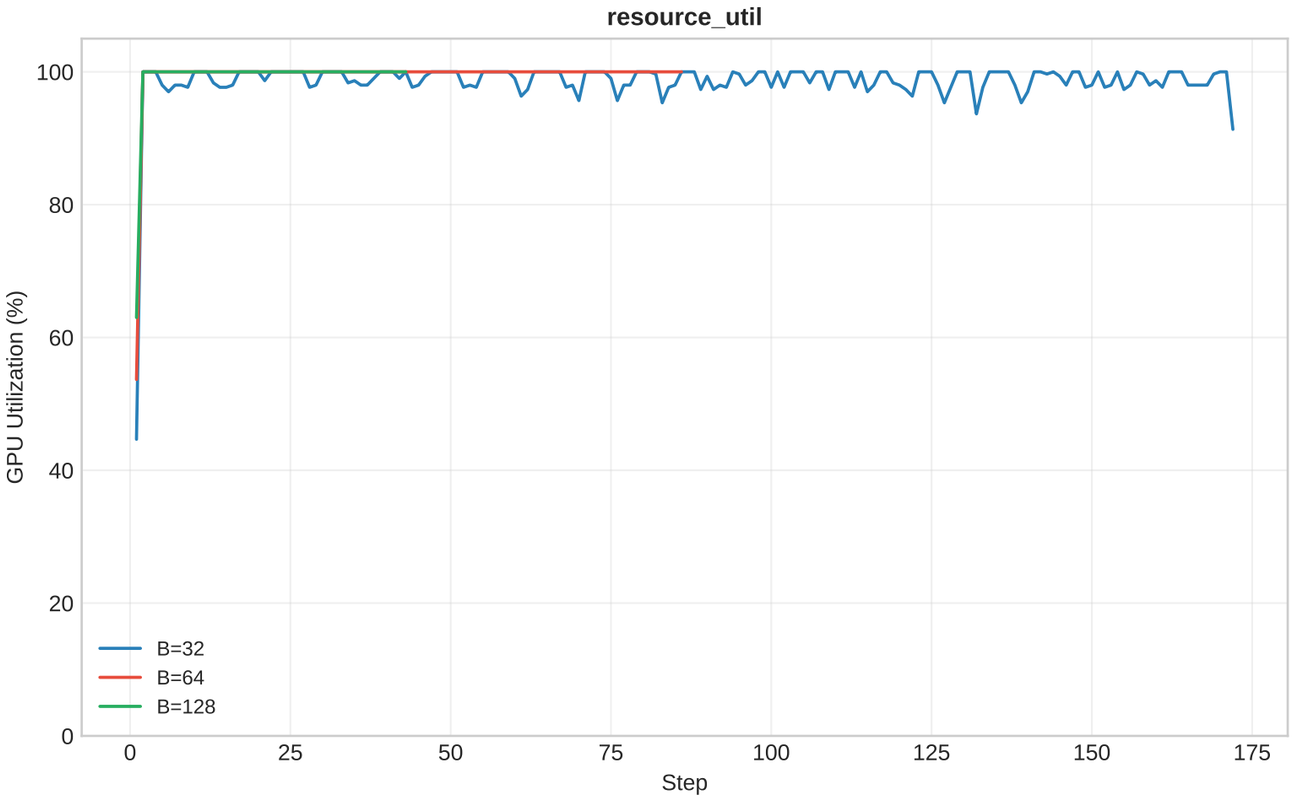
Data	W	B	Trainer	Steps	GPU Util Mean	GPU Util Std	CPU Util Mean	GPU Mem (GB)
Disk	0	32	resource_util	172	96.9%	5.3	132.0%	0.14
Disk	0	32	resource_util_max	172	96.6%	5.1	127.0%	6.81
Disk	0	64	resource_util	86	99.5%	4.6	128.5%	0.14
Disk	0	64	resource_util_max	86	99.4%	5.5	124.3%	12.49
Disk	0	128	resource_util	43	99.2%	5.0	128.5%	0.14
Disk	0	128	resource_util_max	43	99.0%	6.7	122.0%	23.94
Disk	4	32	resource_util	172	98.6%	4.4	98.8%	0.14
Disk	4	32	resource_util_max	172	98.5%	2.6	98.8%	6.81
Disk	4	64	resource_util	86	99.5%	5.0	98.6%	0.14
Disk	4	64	resource_util_max	86	99.8%	2.0	98.6%	12.49
Disk	4	128	resource_util	43	99.1%	5.6	95.3%	0.14
Disk	4	128	resource_util_max	43	99.6%	2.5	96.6%	23.94
Milabench	0	32	resource_util	500	99.6%	0.7	110.1%	0.14
Milabench	0	32	resource_util_max	500	99.6%	0.7	109.9%	6.81
Milabench	0	64	resource_util	250	100.0%	0.4	109.7%	0.14
Milabench	0	64	resource_util_max	250	100.0%	0.4	110.1%	12.49
Milabench	0	128	resource_util	125	100.0%	0.1	107.1%	0.14
Milabench	0	128	resource_util_max	125	100.0%	0.5	107.5%	23.94
Milabench	4	32	resource_util	500	99.1%	1.4	98.7%	0.14
Milabench	4	32	resource_util_max	500	99.1%	1.3	99.7%	6.81
Milabench	4	64	resource_util	250	100.0%	0.2	99.7%	0.14
Milabench	4	64	resource_util_max	250	100.0%	0.0	99.8%	12.49
Milabench	4	128	resource_util	125	100.0%	0.1	99.6%	0.14
Milabench	4	128	resource_util_max	125	100.0%	0.0	99.5%	23.94

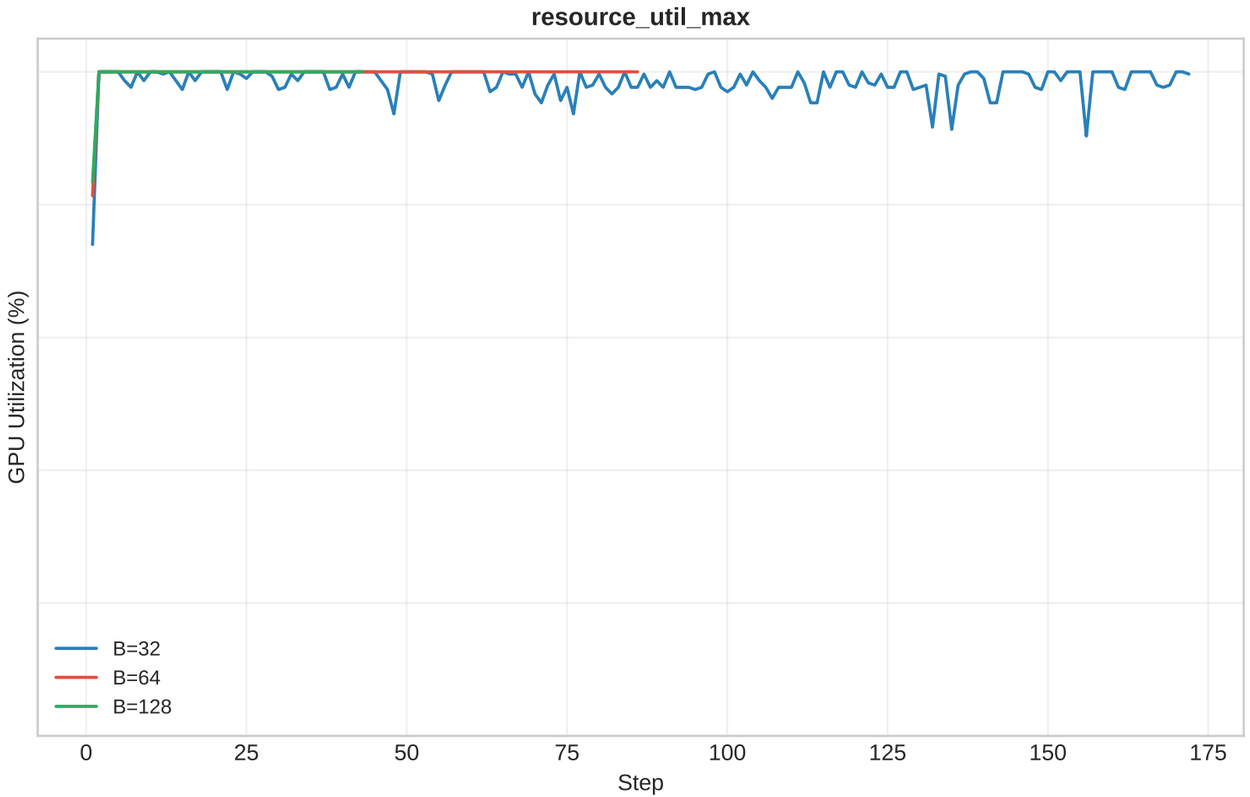
GPU Utilization Across Batch Sizes — Disk, w=0



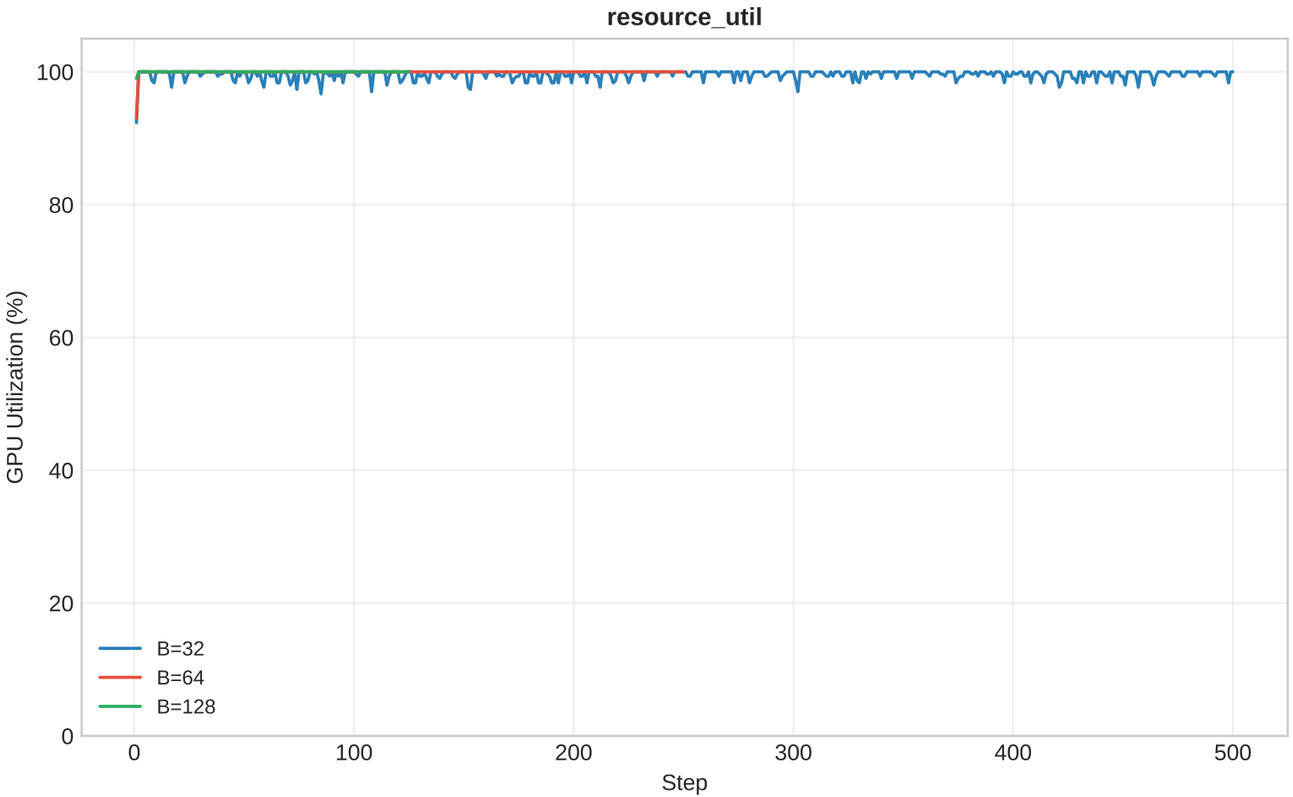


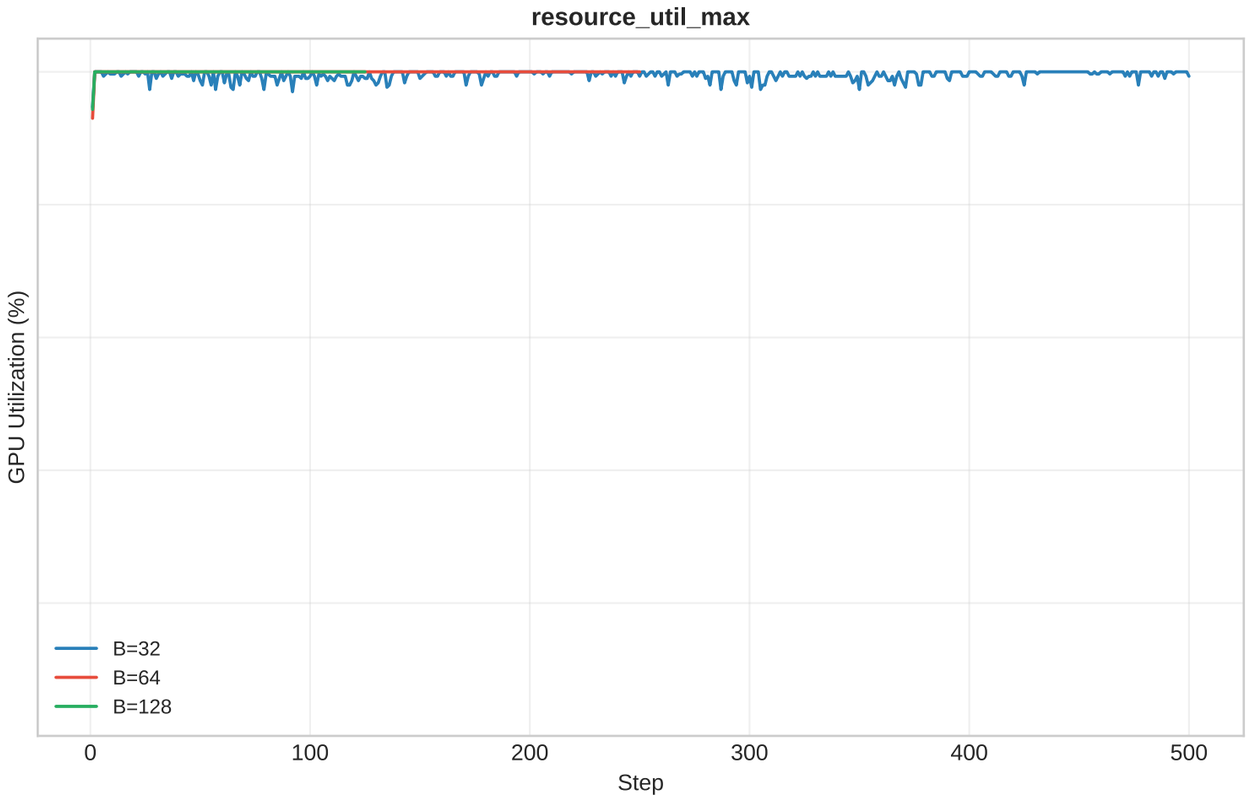
GPU Utilization Across Batch Sizes — Disk, w=4



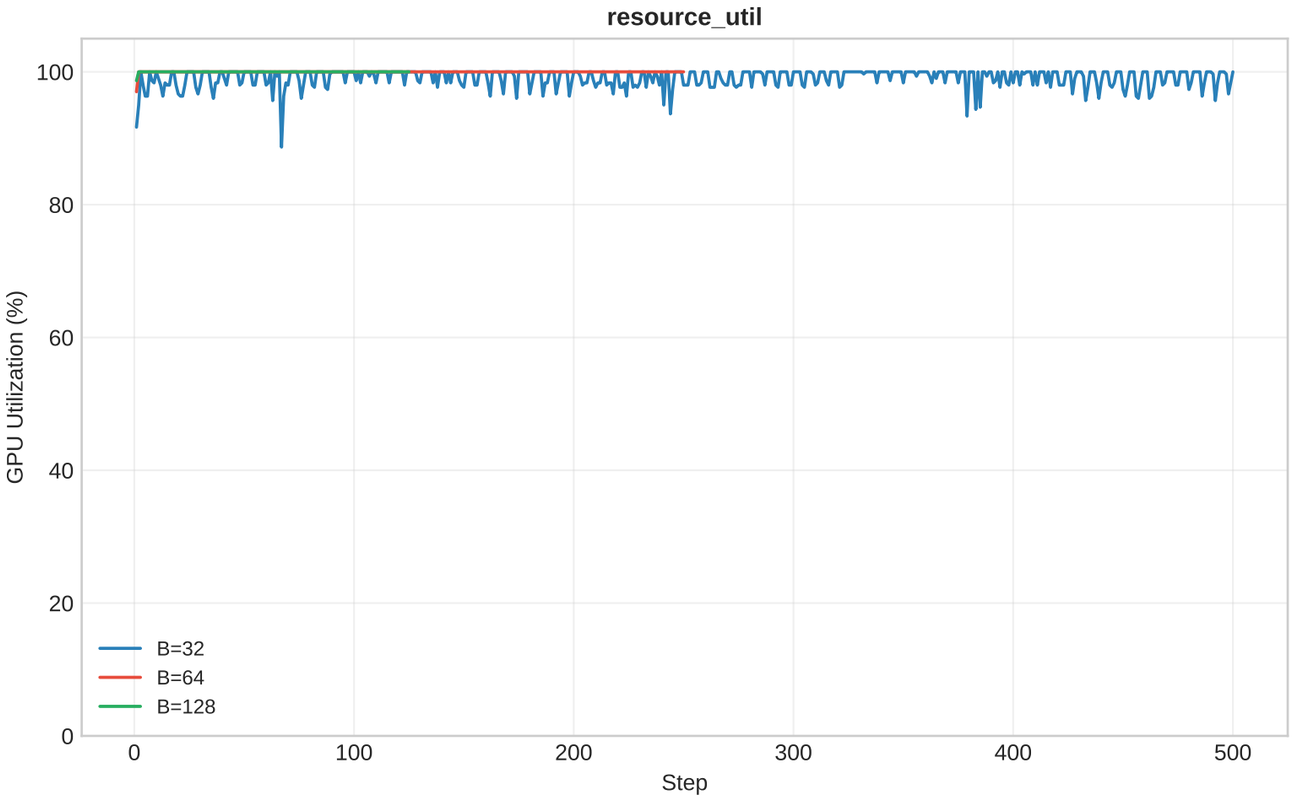


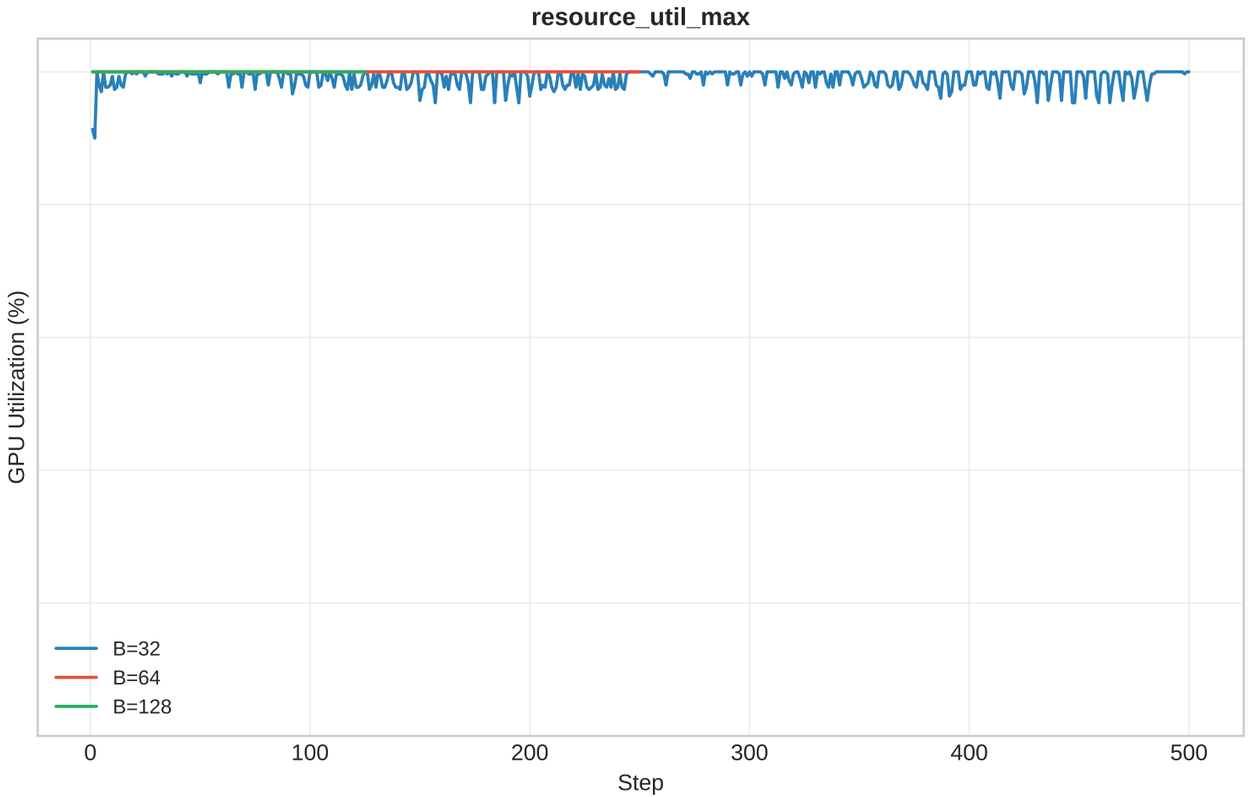
GPU Utilization Across Batch Sizes — Milabench, w=0



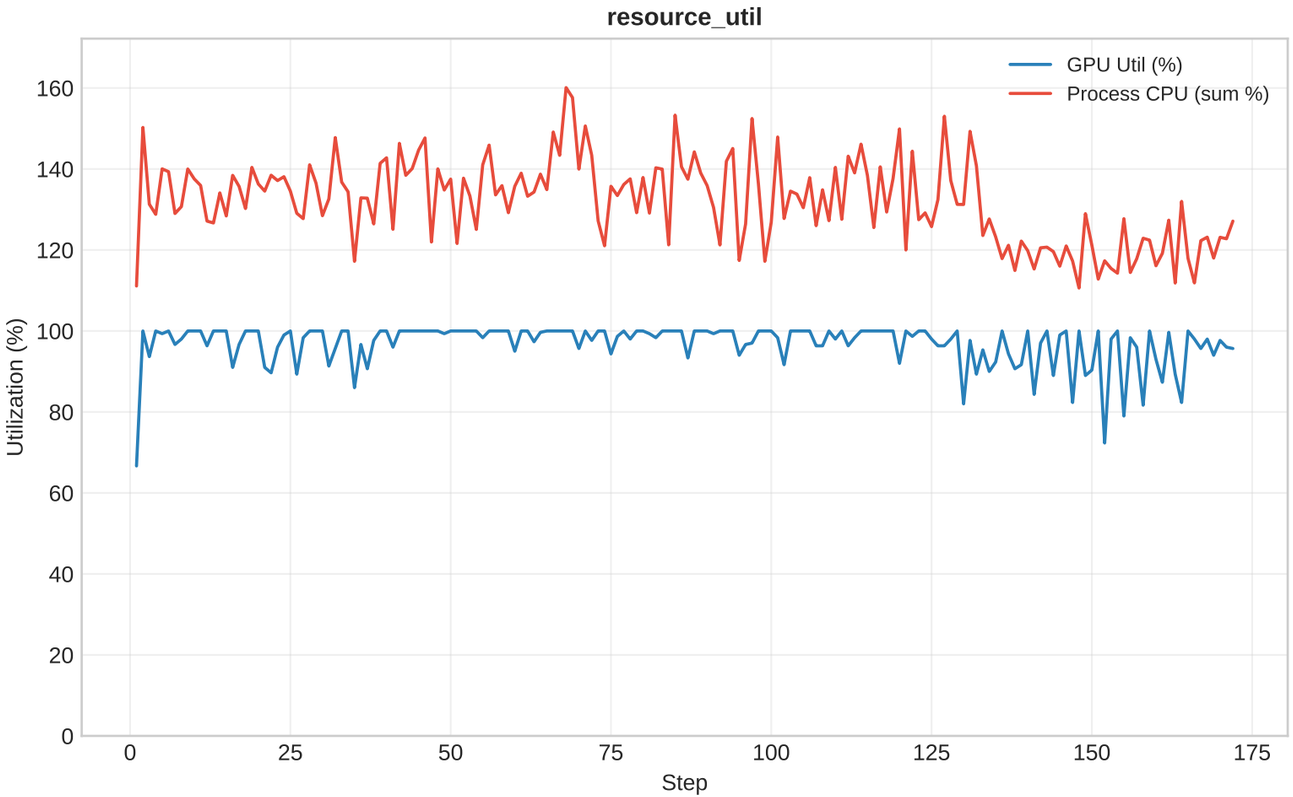


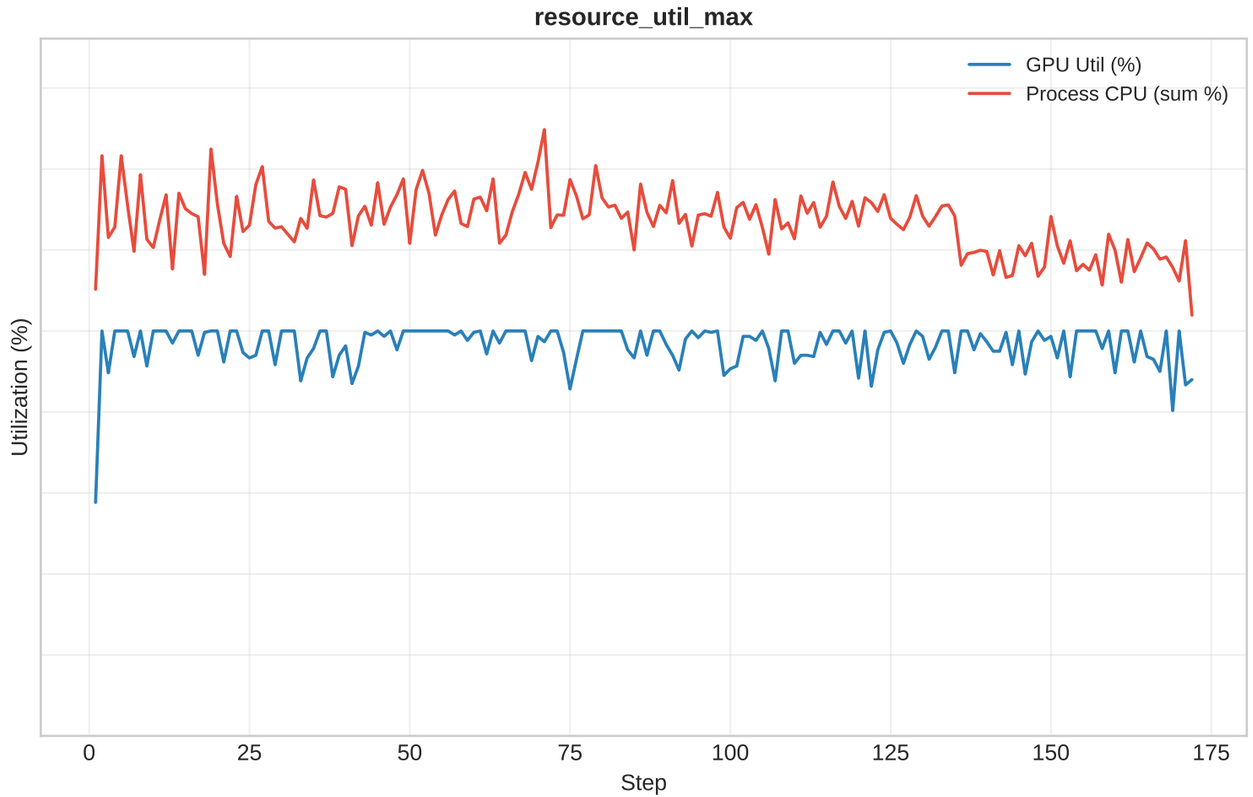
GPU Utilization Across Batch Sizes — Milabench, w=4



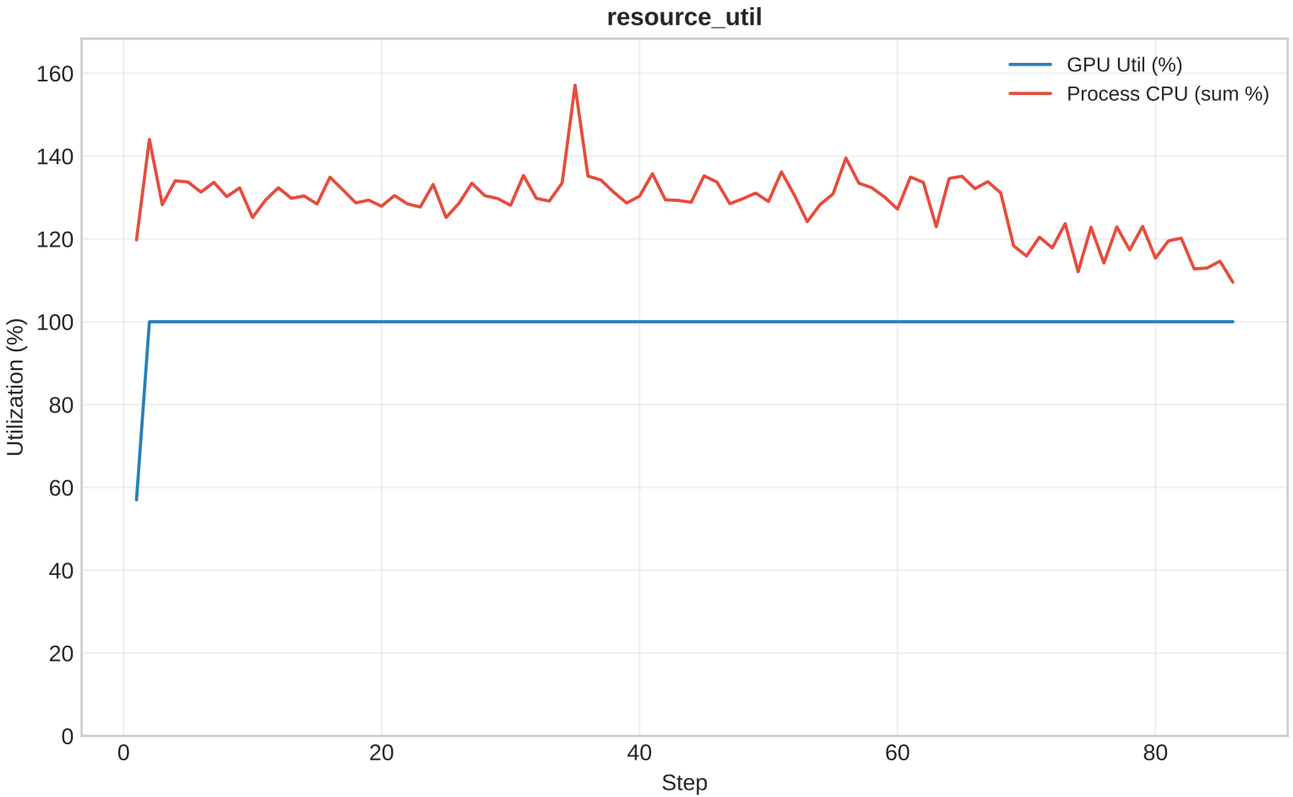


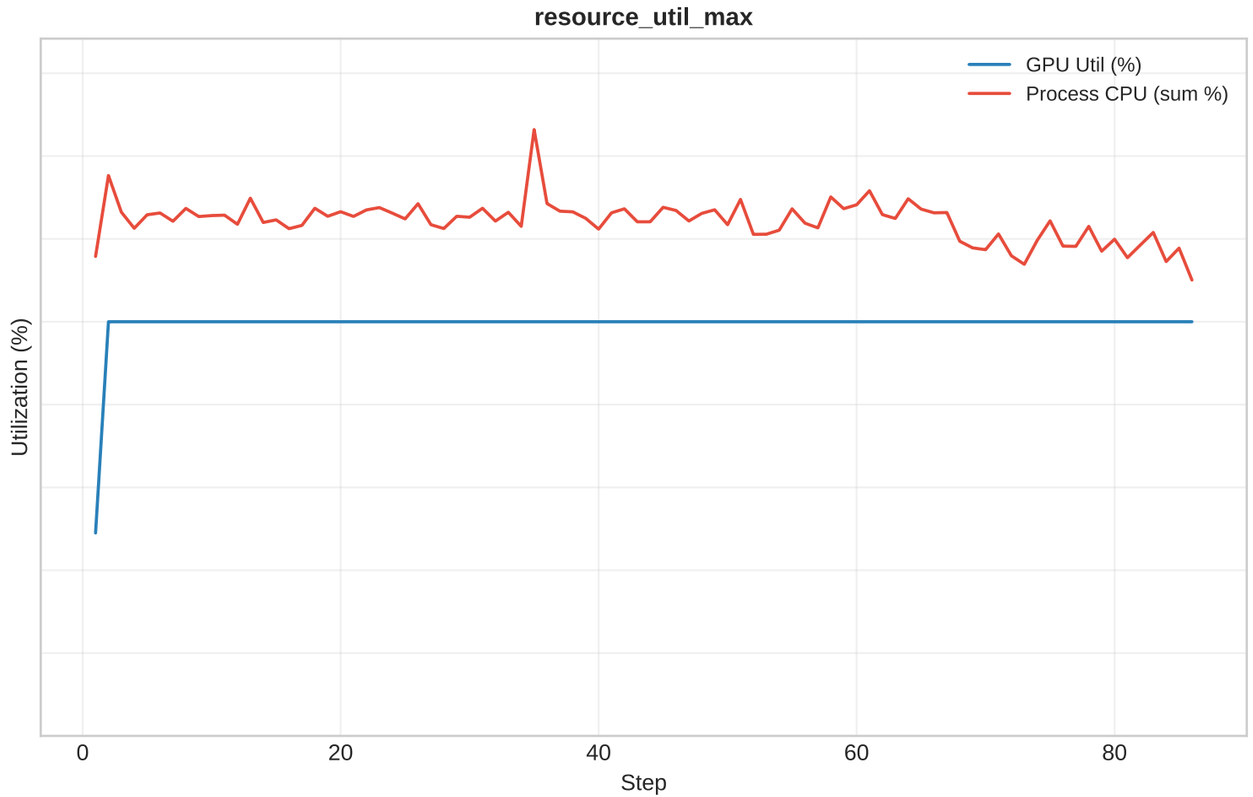
GPU vs CPU Utilization — Disk, w=0, B=32



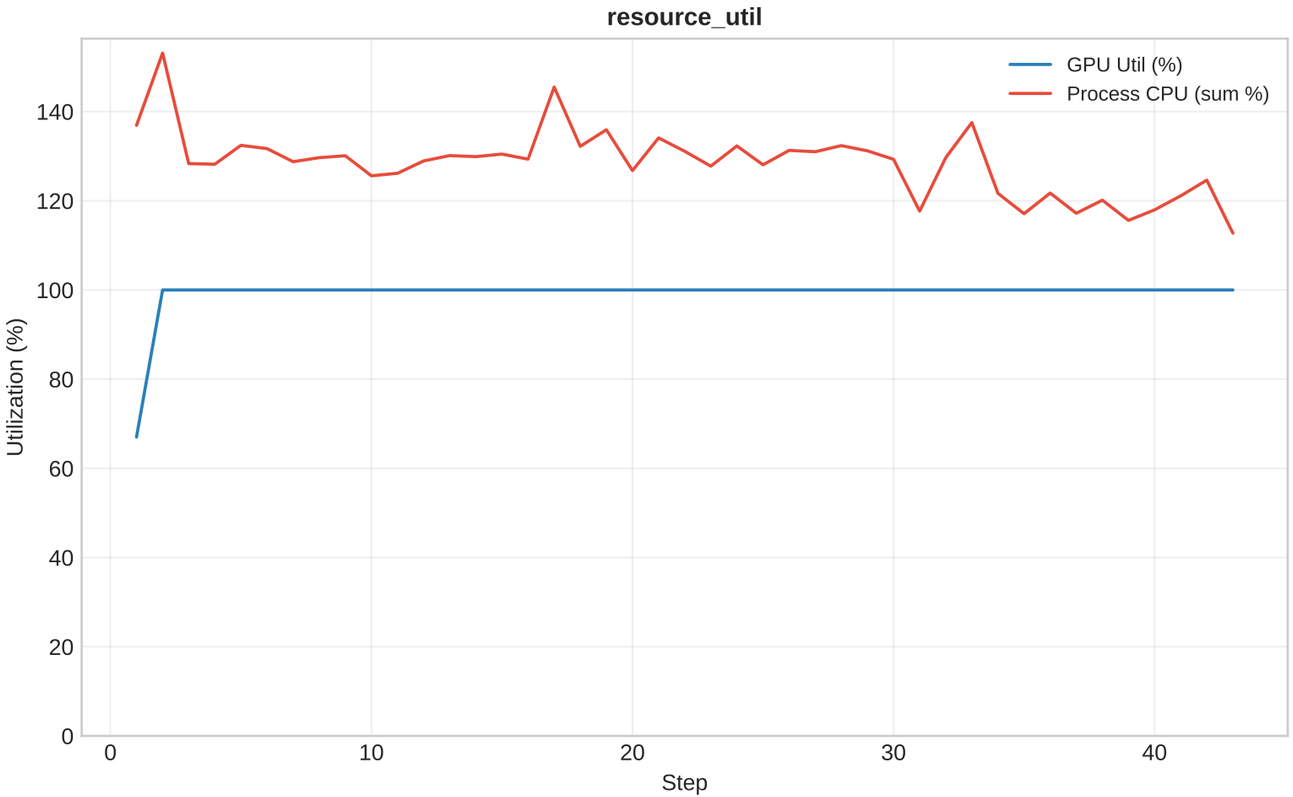


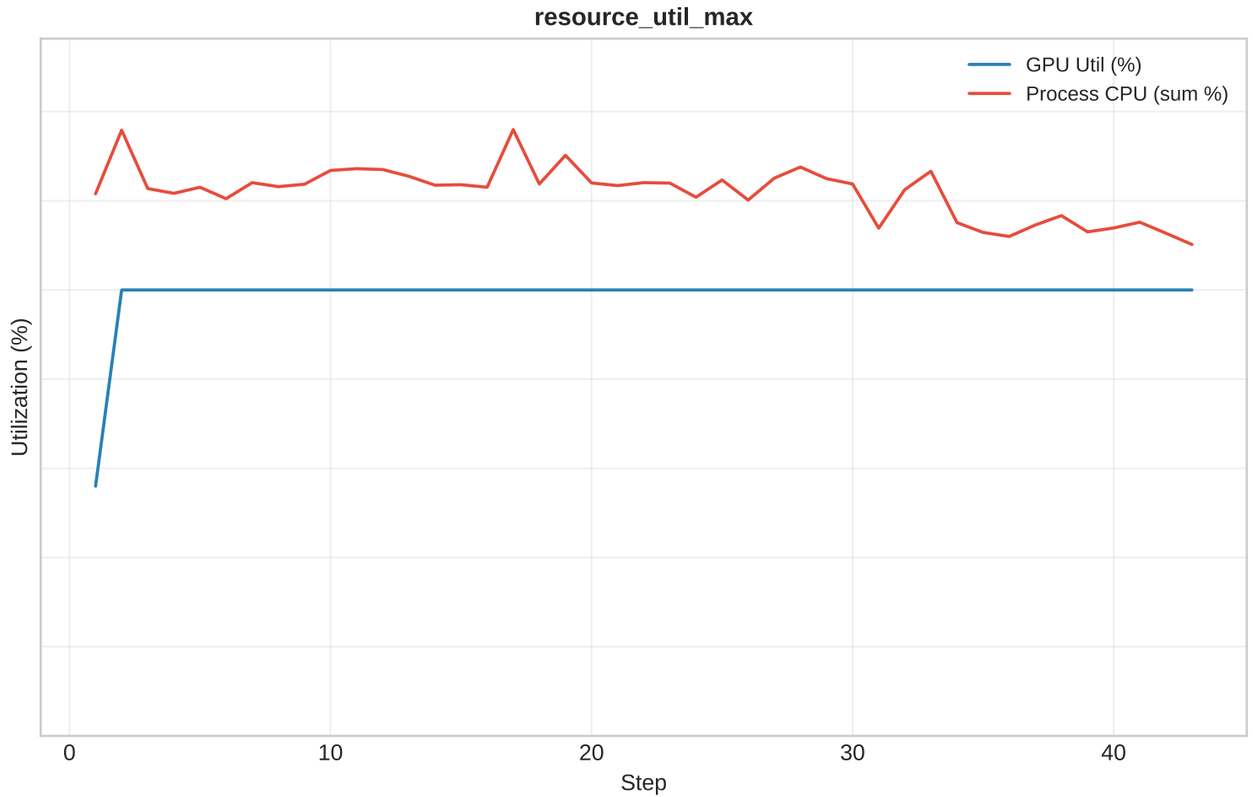
GPU vs CPU Utilization — Disk, w=0, B=64



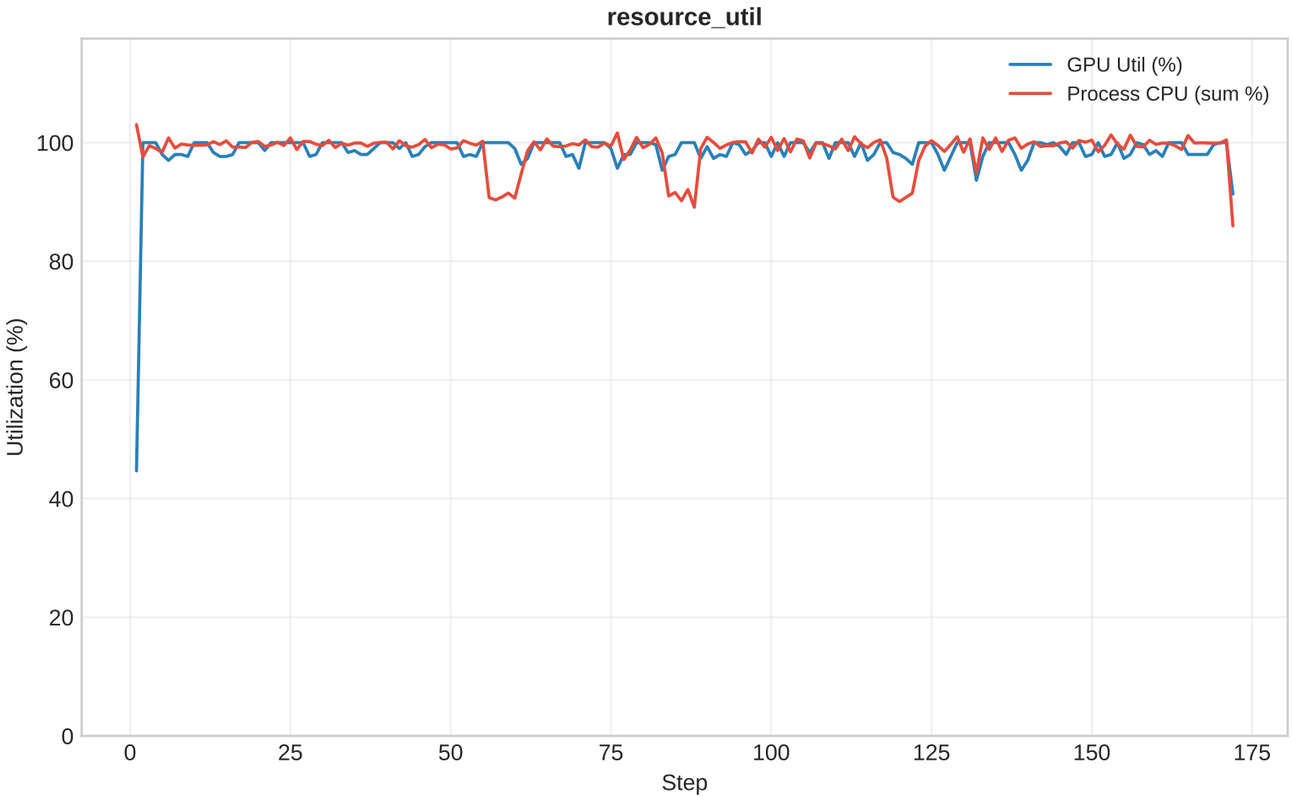


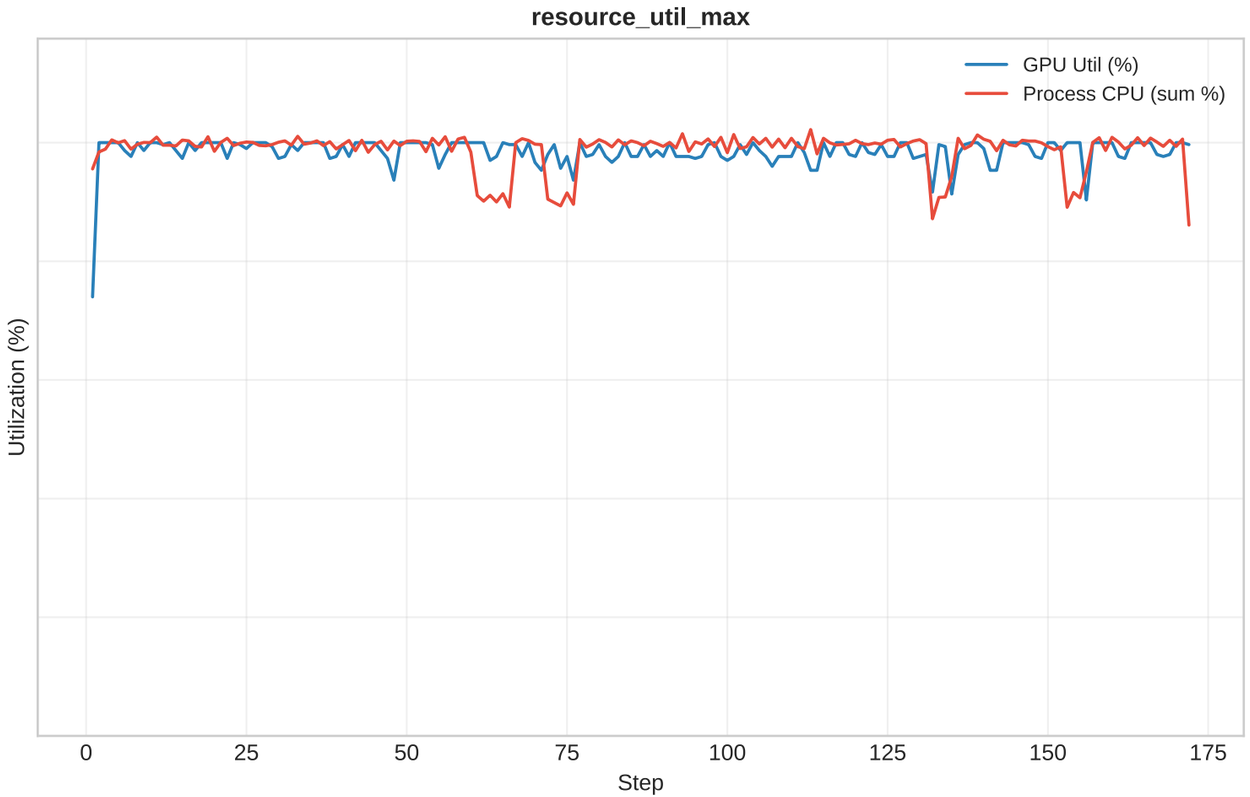
GPU vs CPU Utilization — Disk, w=0, B=128



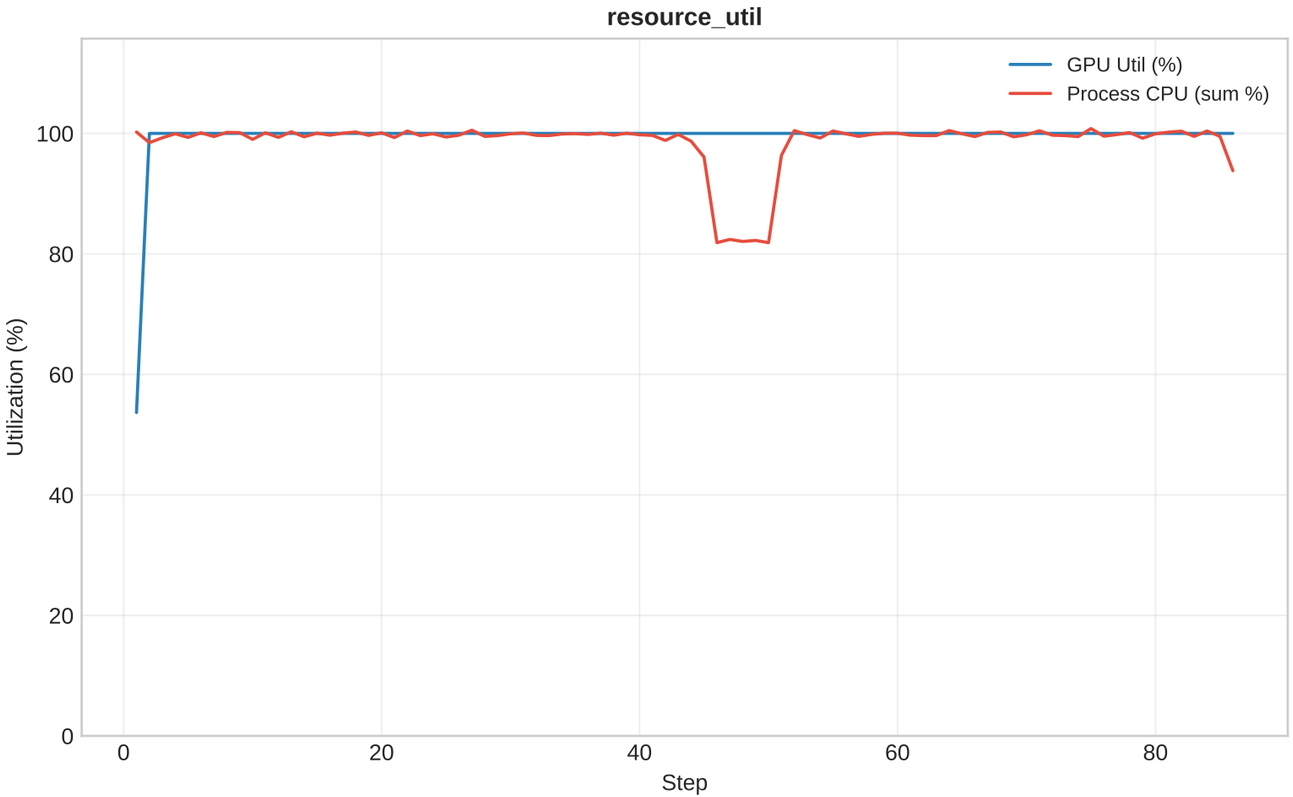


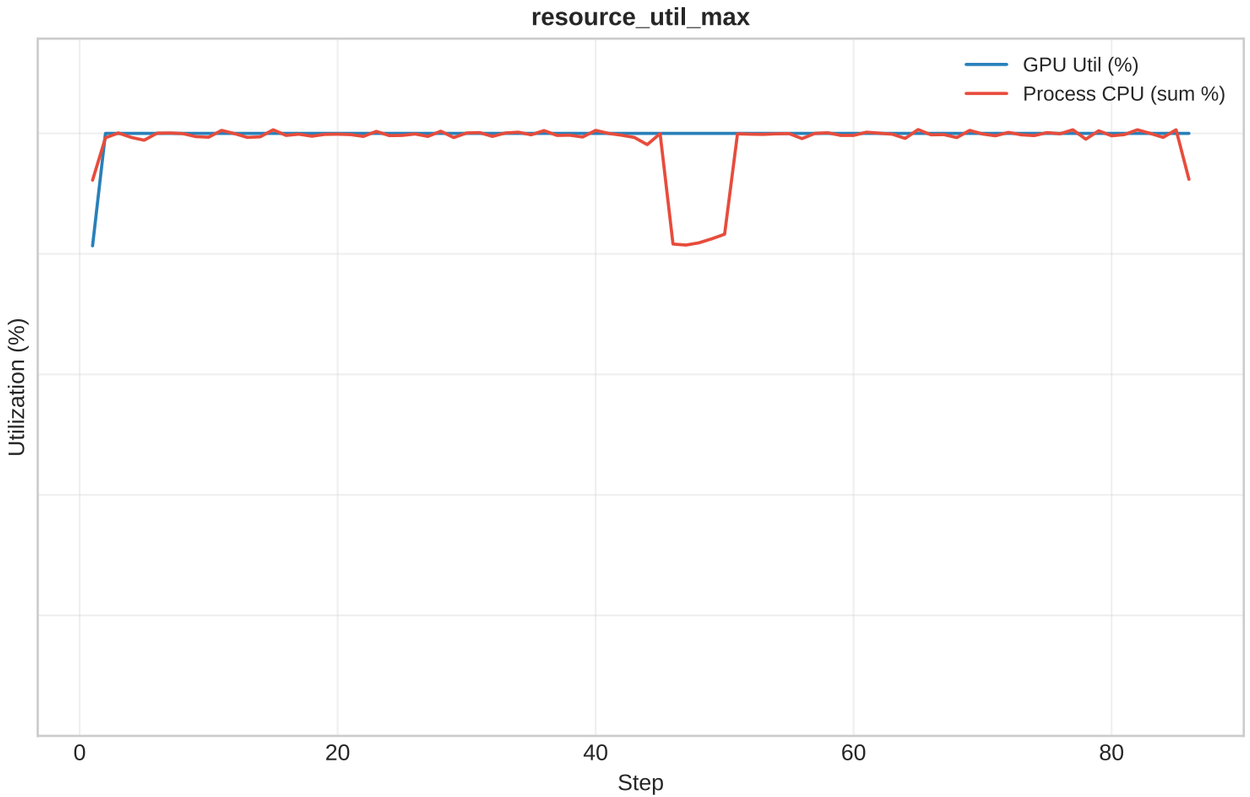
GPU vs CPU Utilization — Disk, w=4, B=32



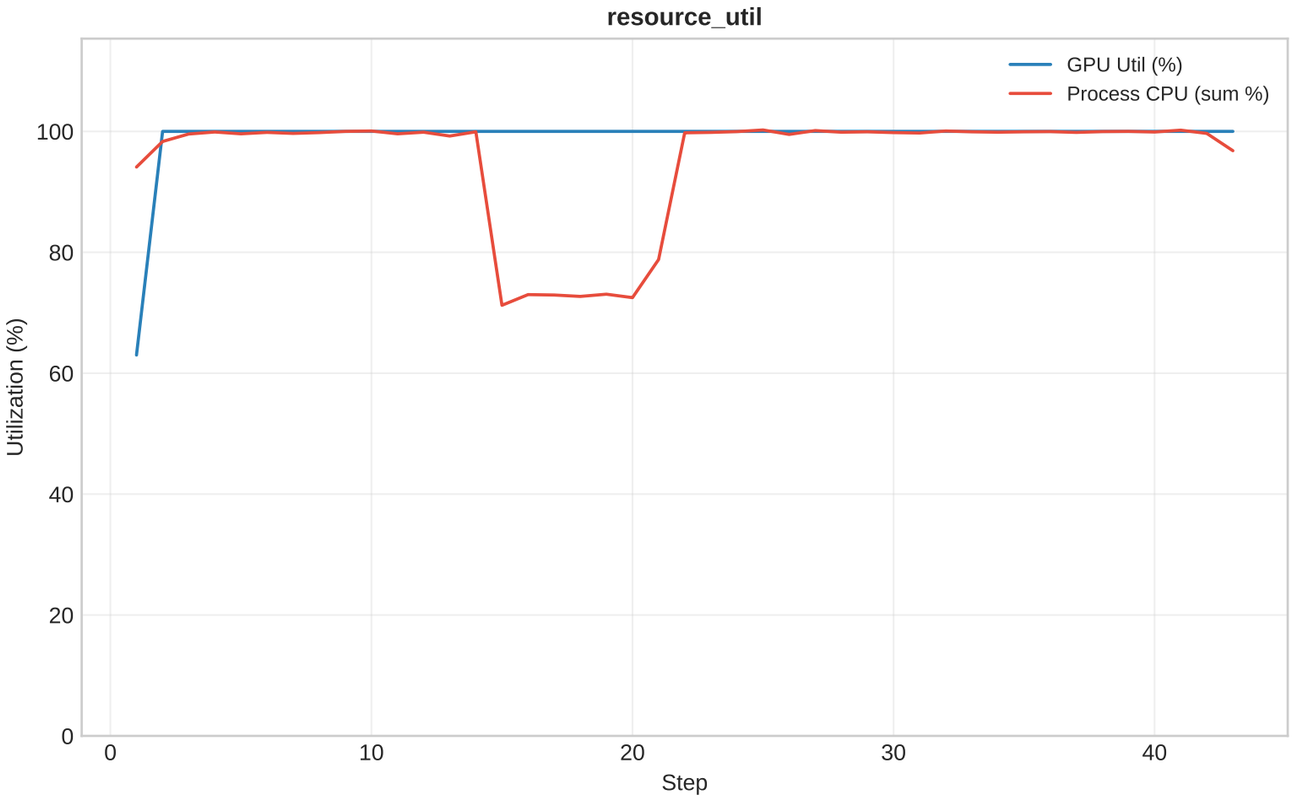


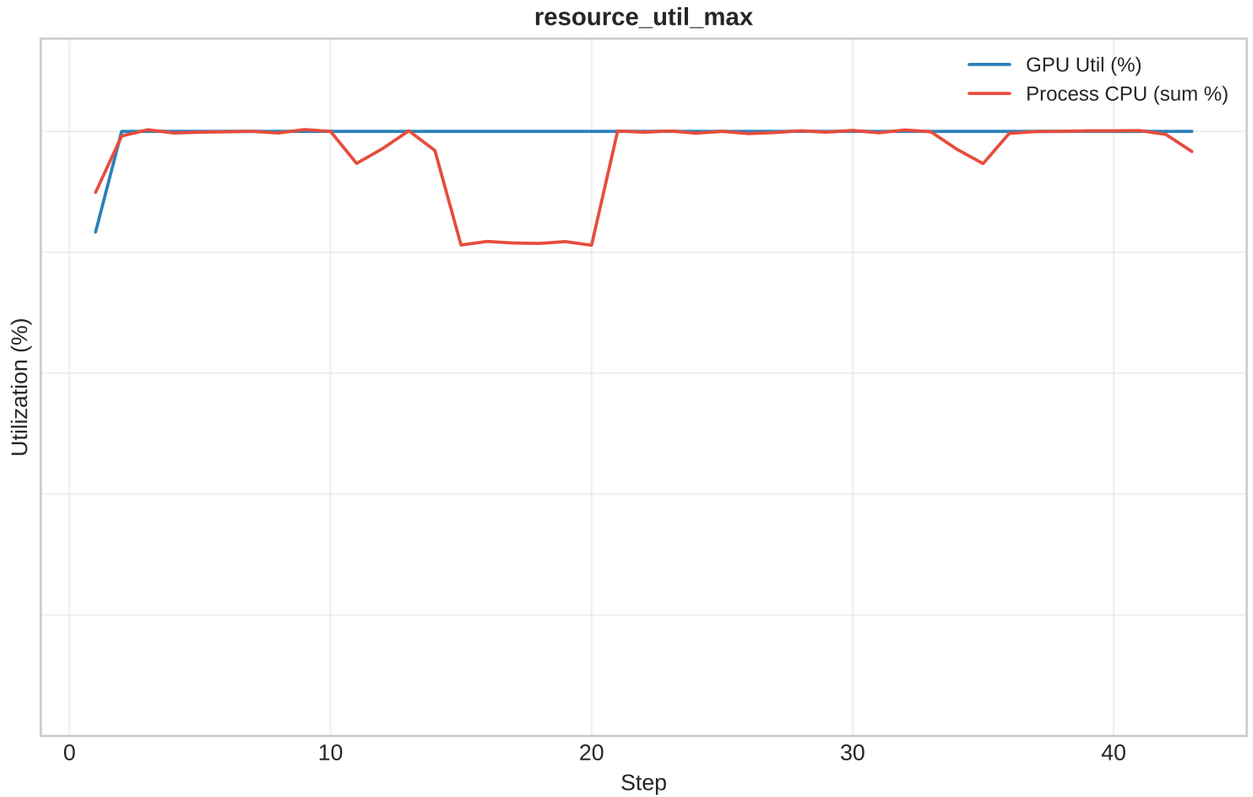
GPU vs CPU Utilization — Disk, w=4, B=64



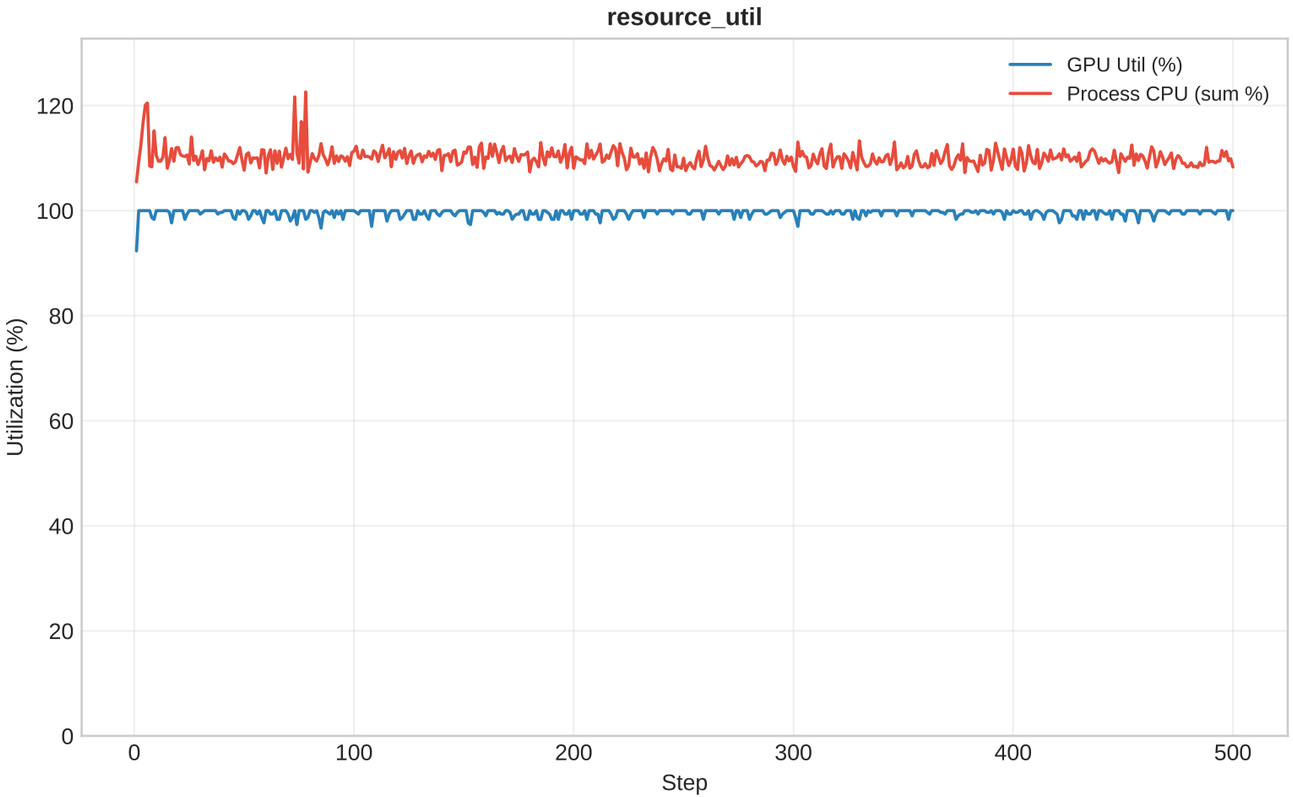


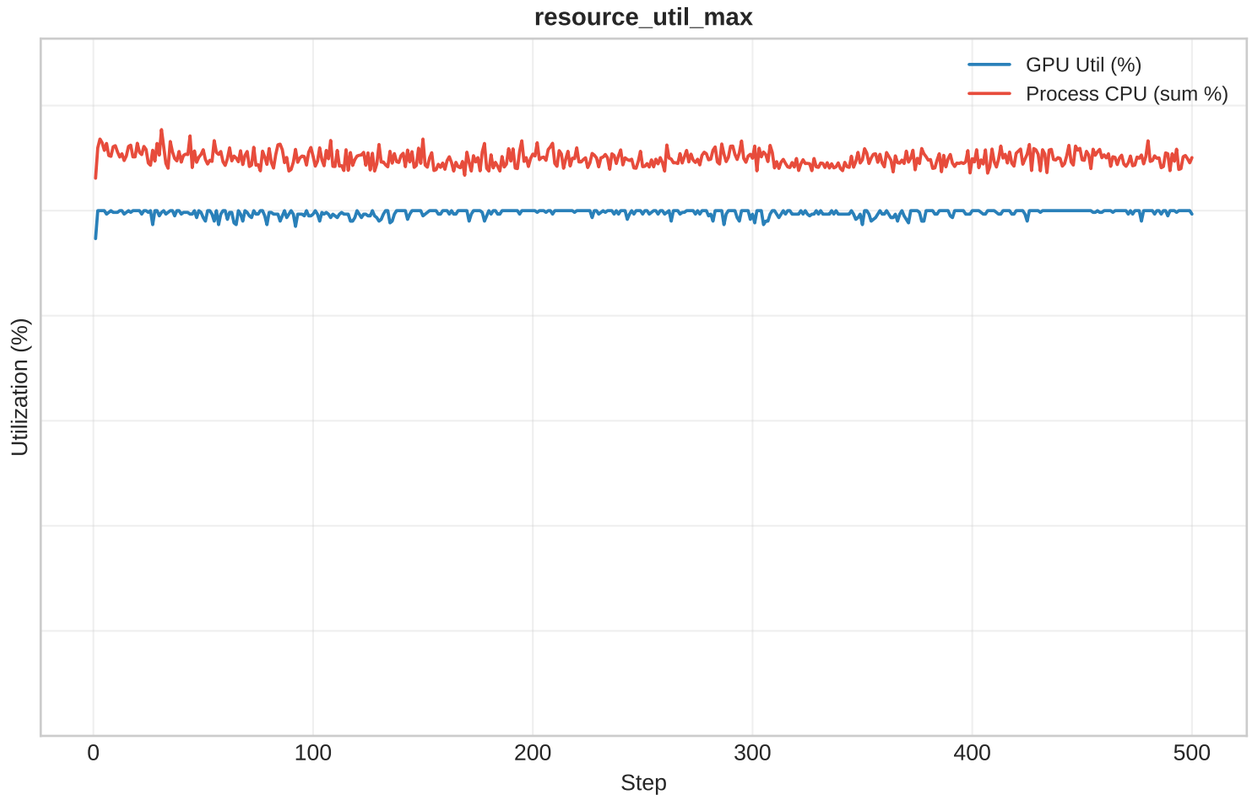
GPU vs CPU Utilization — Disk, w=4, B=128



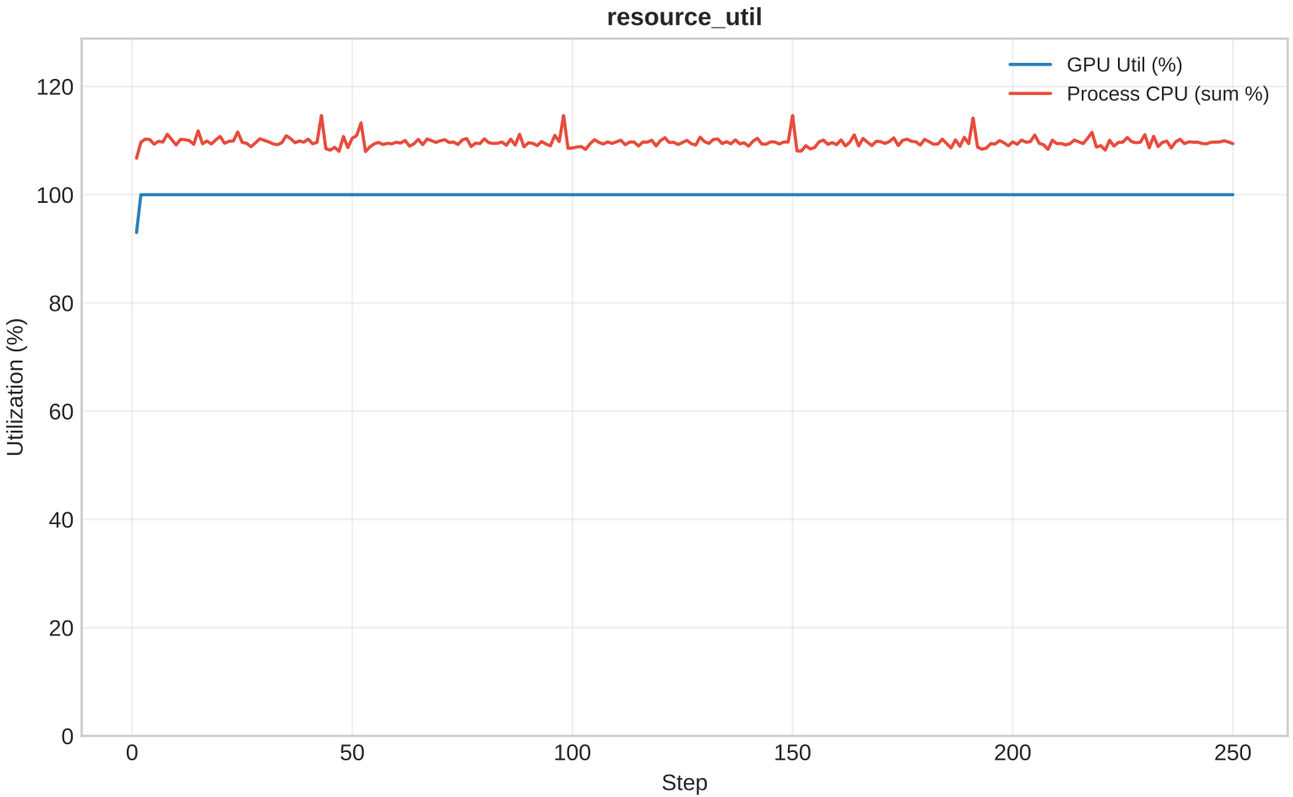


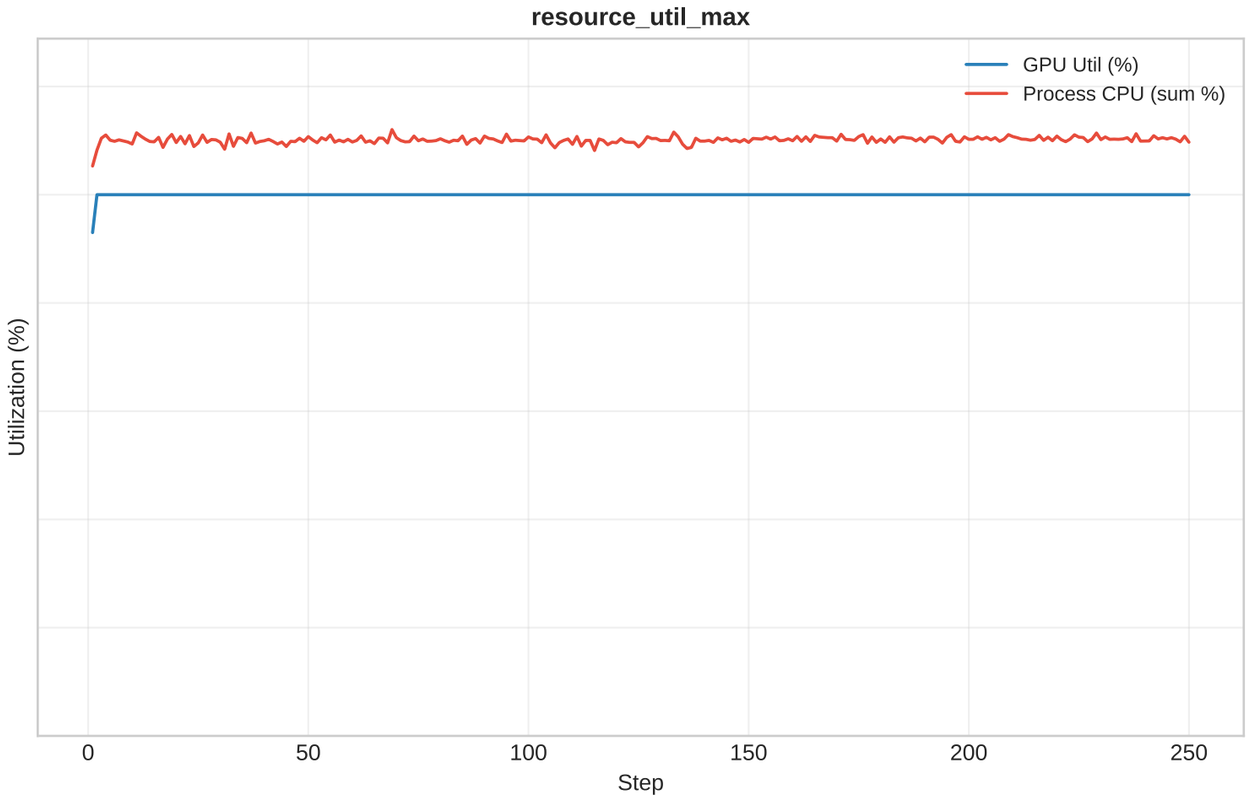
GPU vs CPU Utilization — Milabench, w=0, B=32

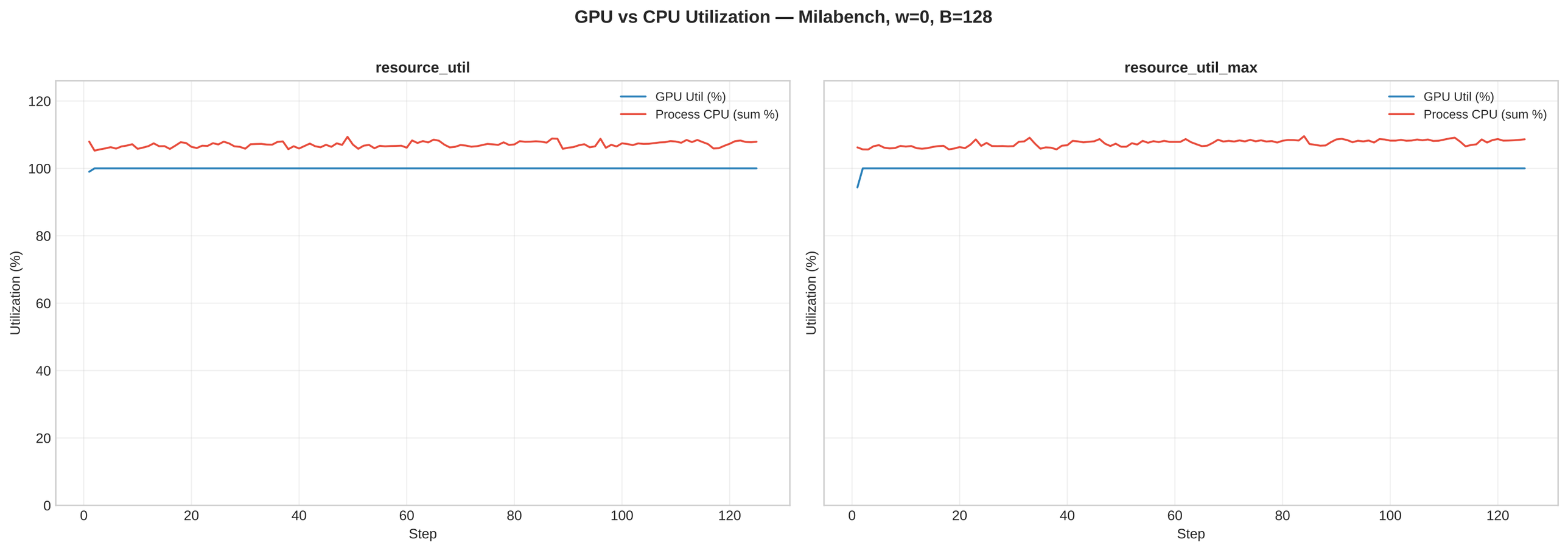


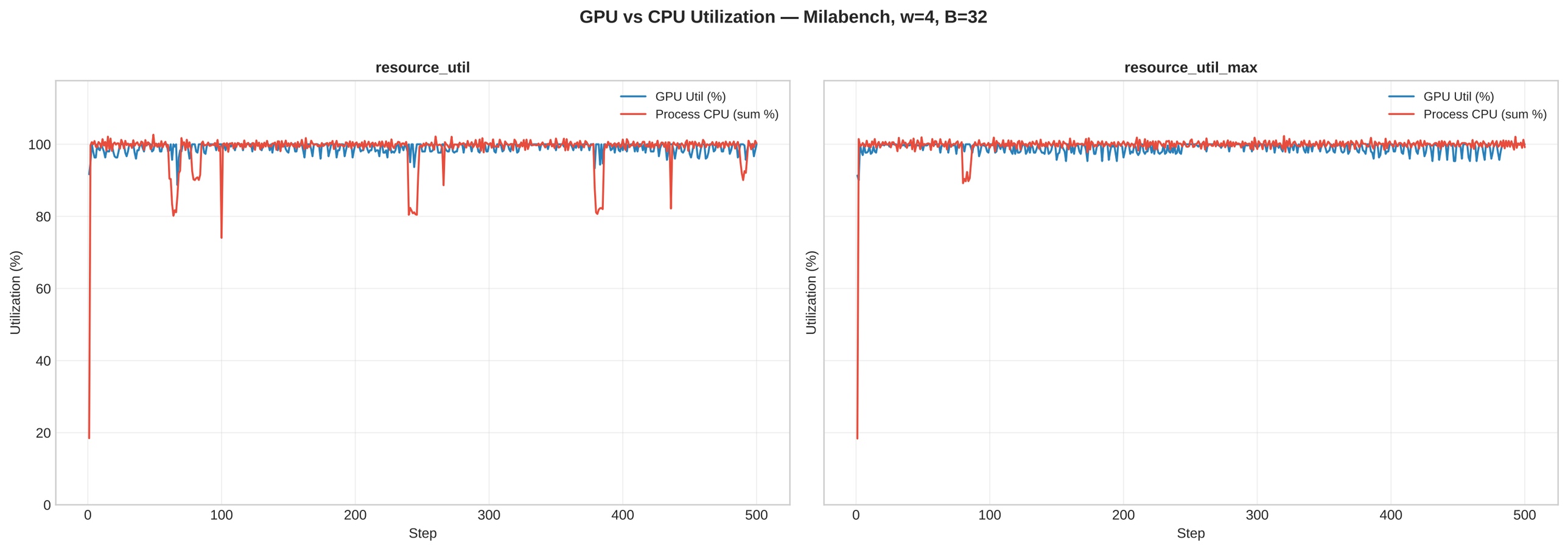


GPU vs CPU Utilization — Milabench, w=0, B=64

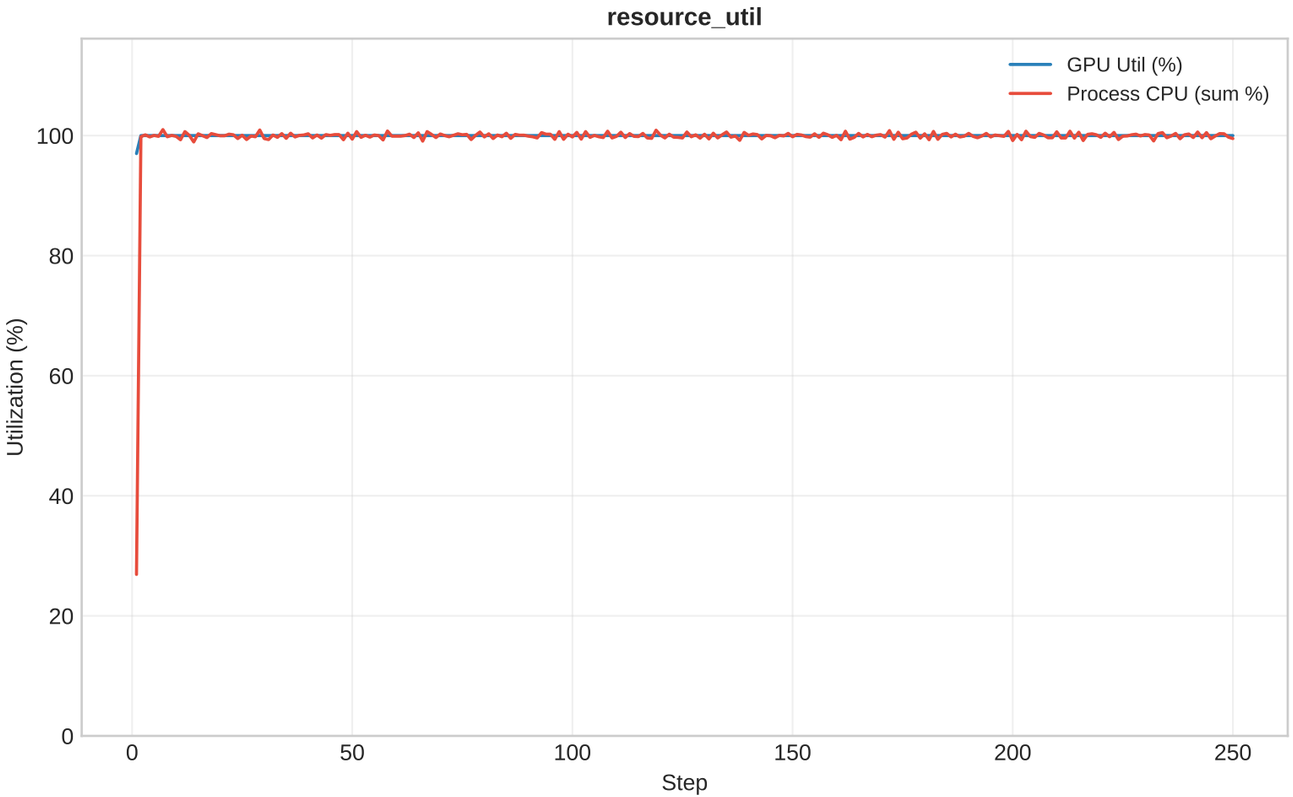


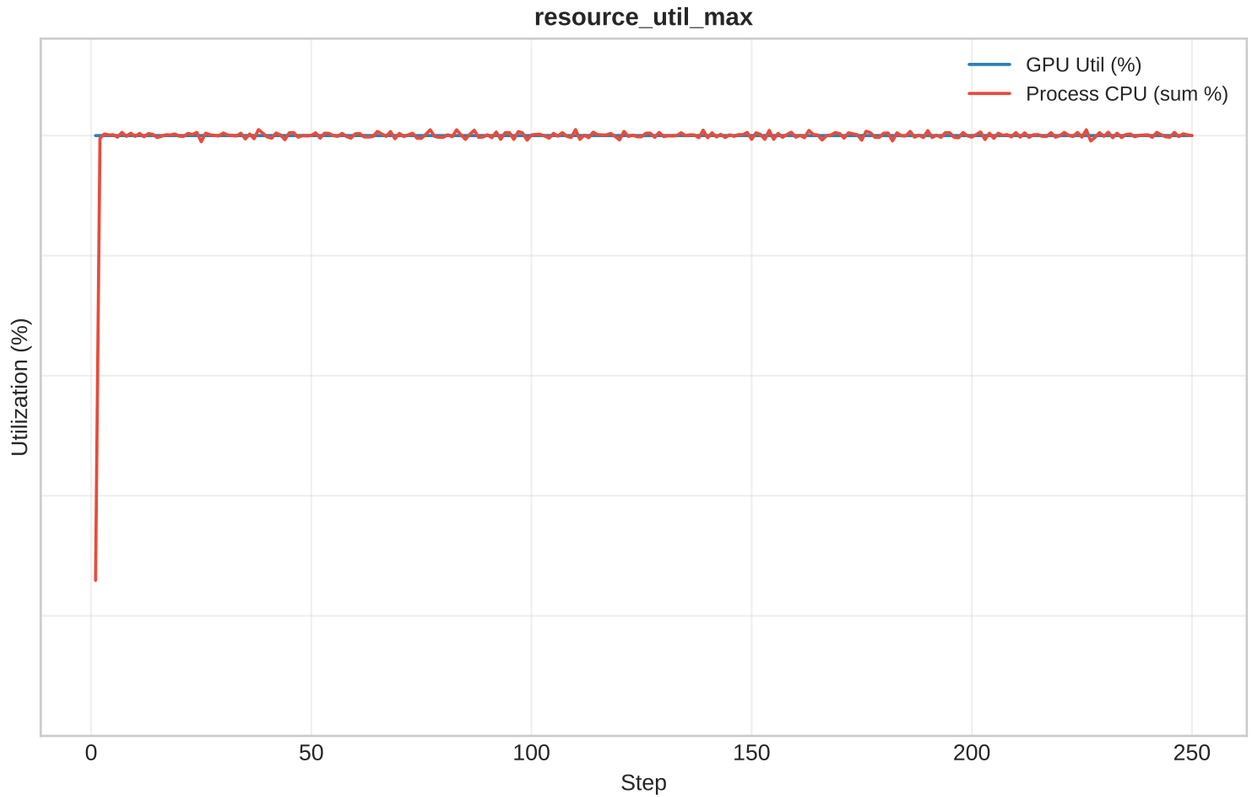


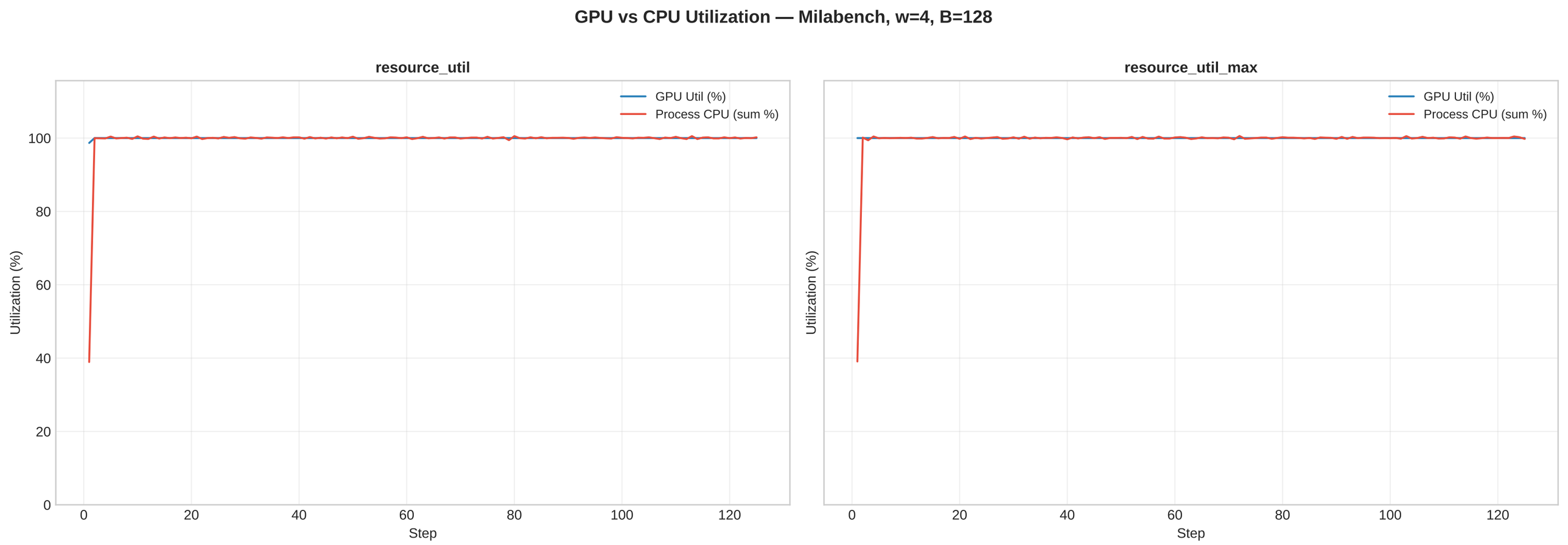

GPU vs CPU Utilization — Milabench, w=0, B=128
resource_util
resource_util_max
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100
120


GPU vs CPU Utilization — Milabench, w=4, B=32
resource_util
resource_util_max
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100
0
100
200
300
400
500

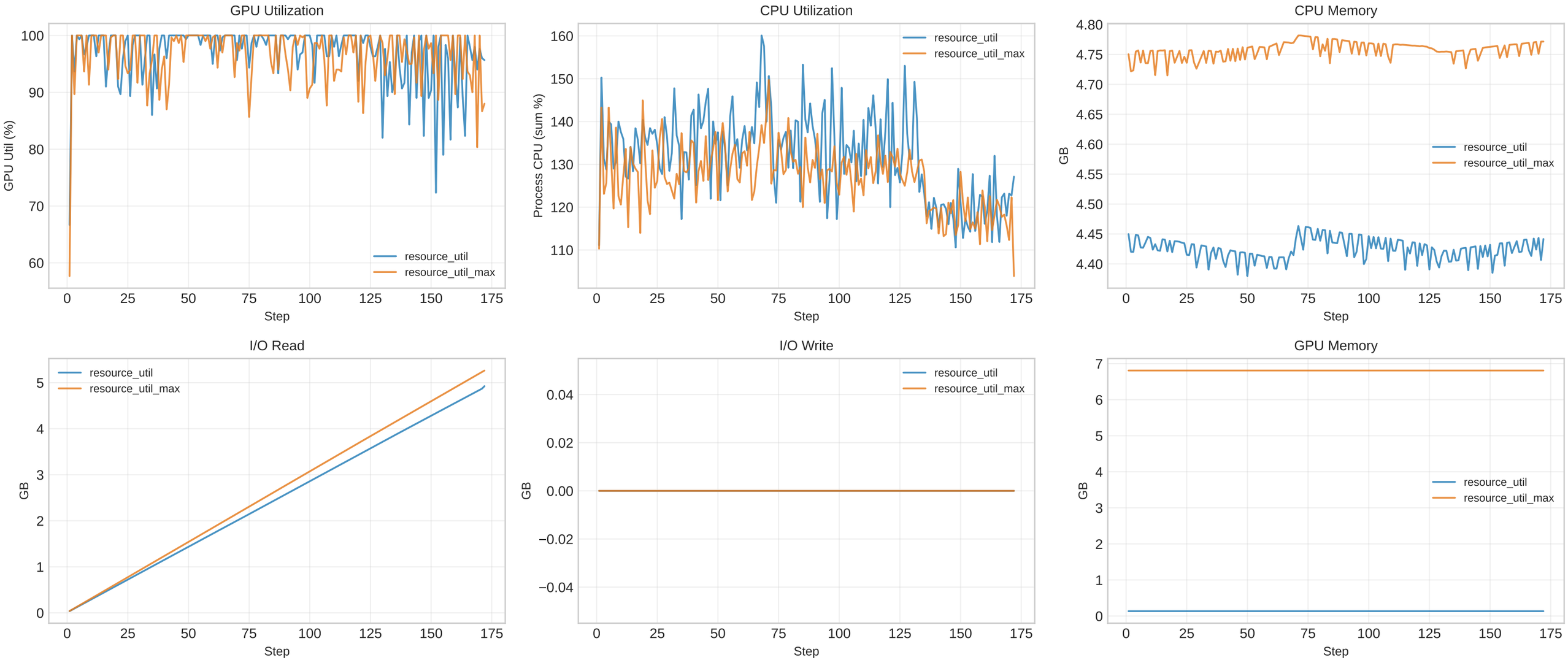
GPU vs CPU Utilization — Milabench, w=4, B=64



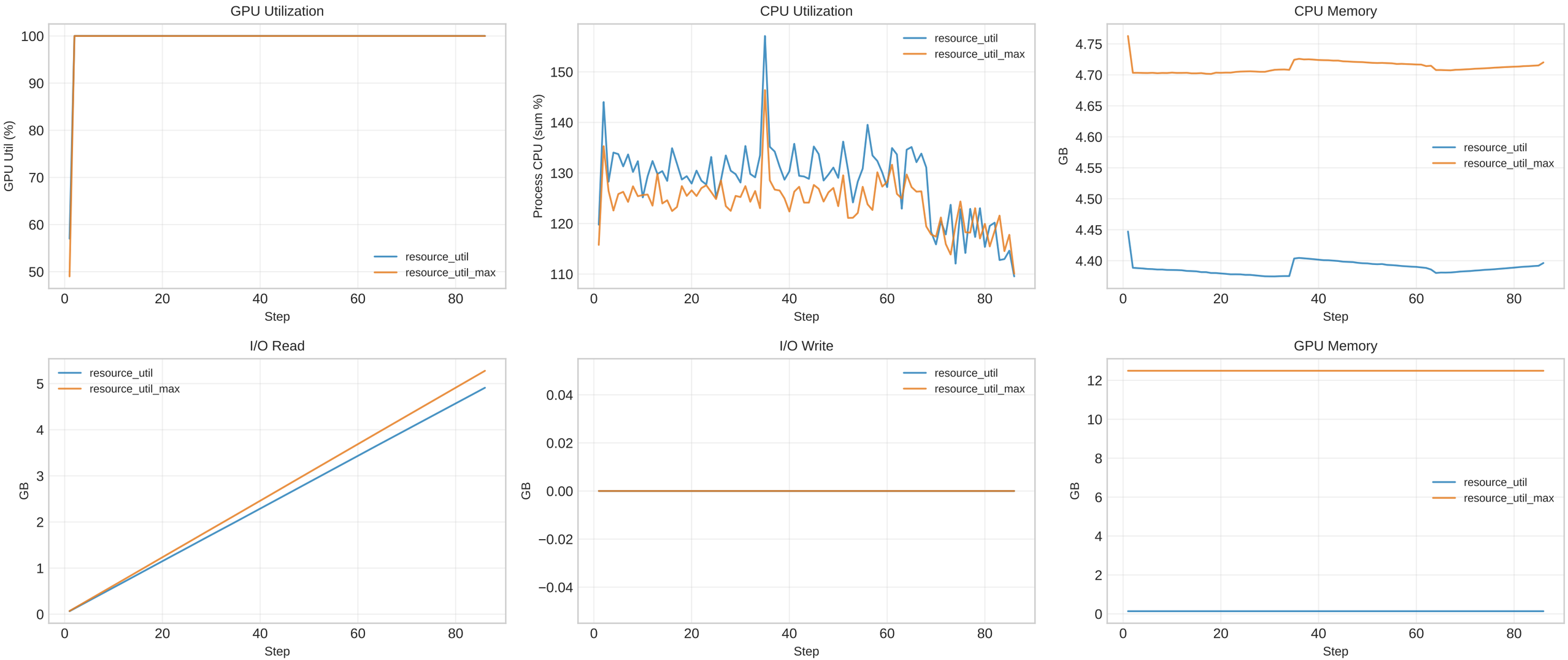



GPU vs CPU Utilization — Milabench, w=4, B=128
resource_util
resource_util_max
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100
120

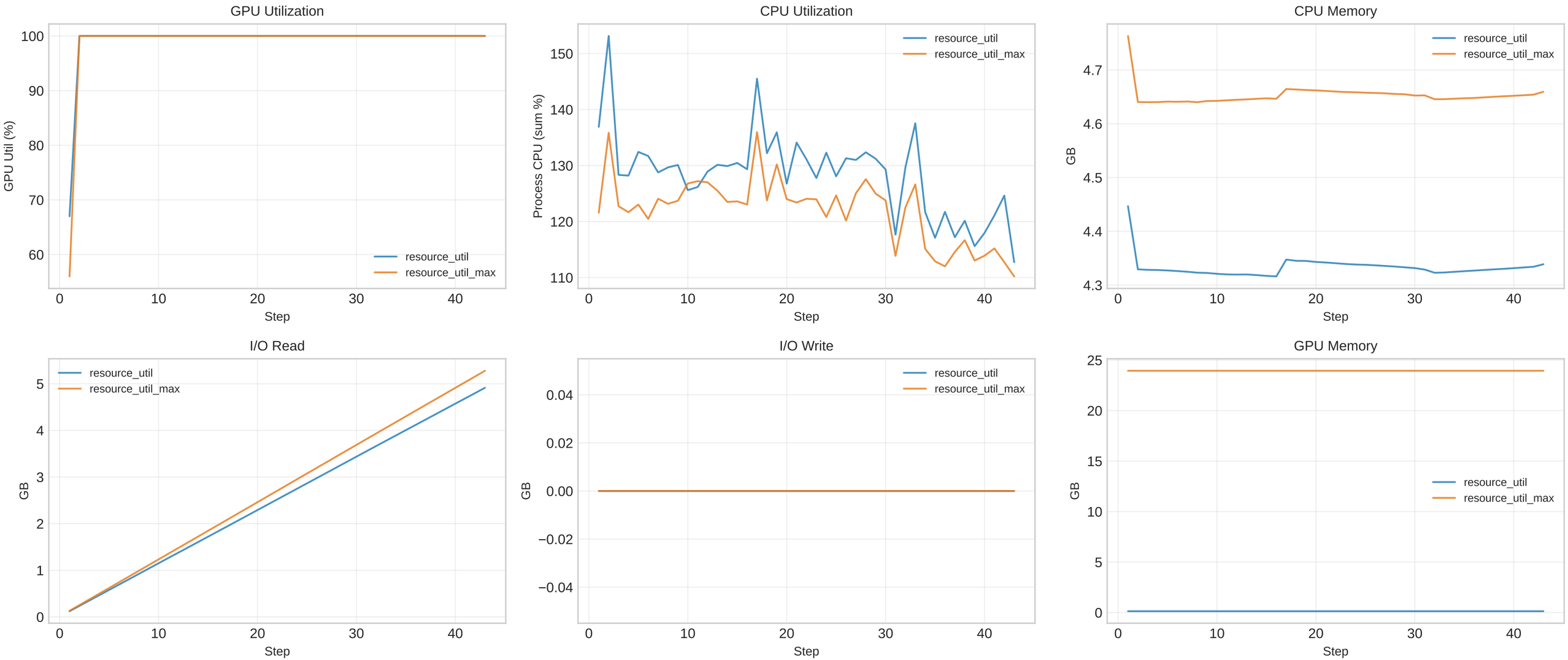
Metric Overlay — Disk, w=0, B=32
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



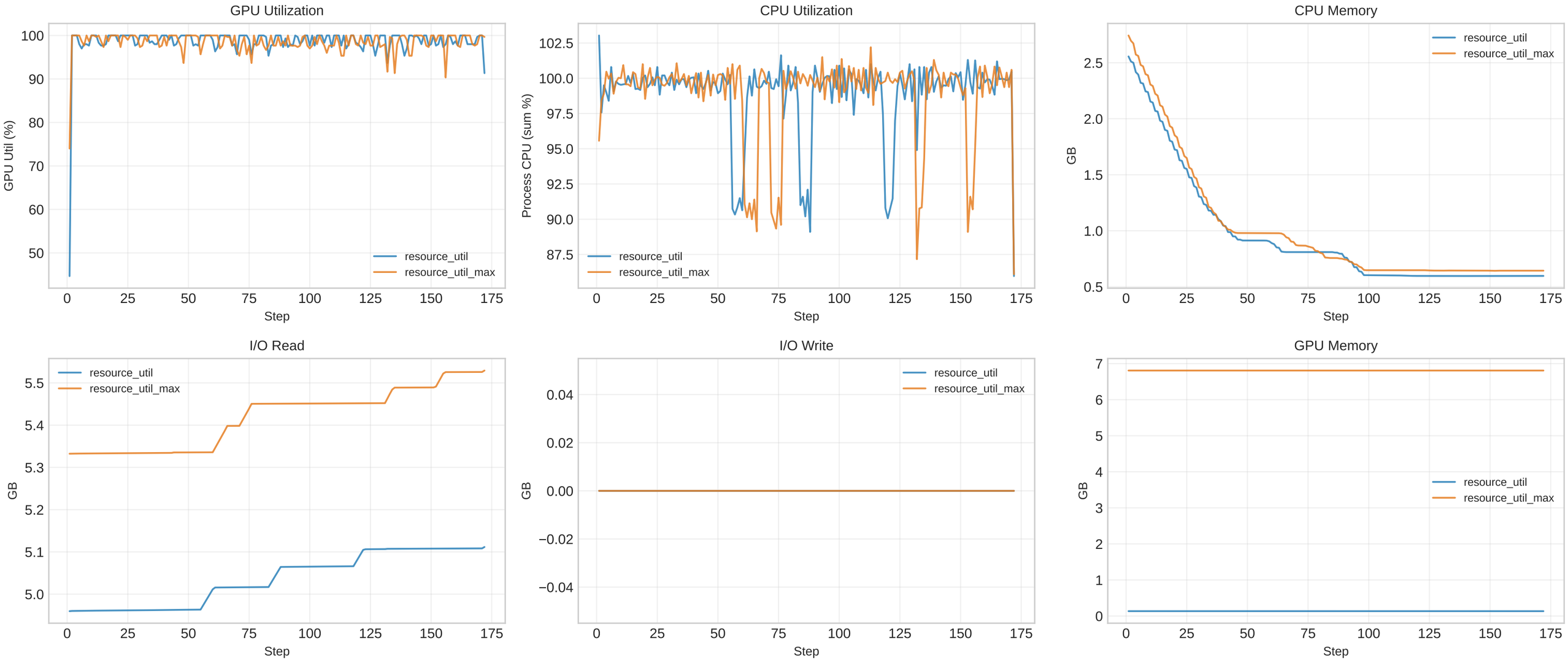
Metric Overlay — Disk, w=0, B=64
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



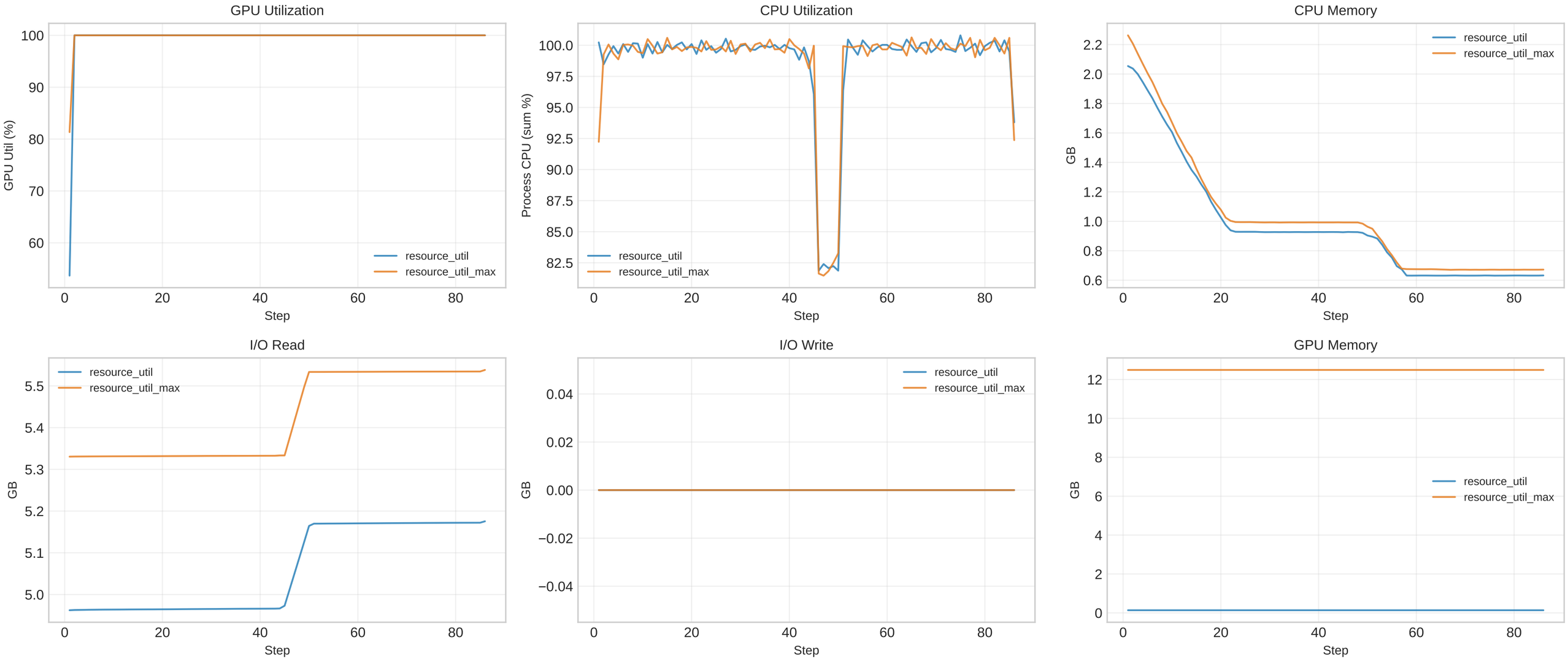
Metric Overlay — Disk, w=0, B=128
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



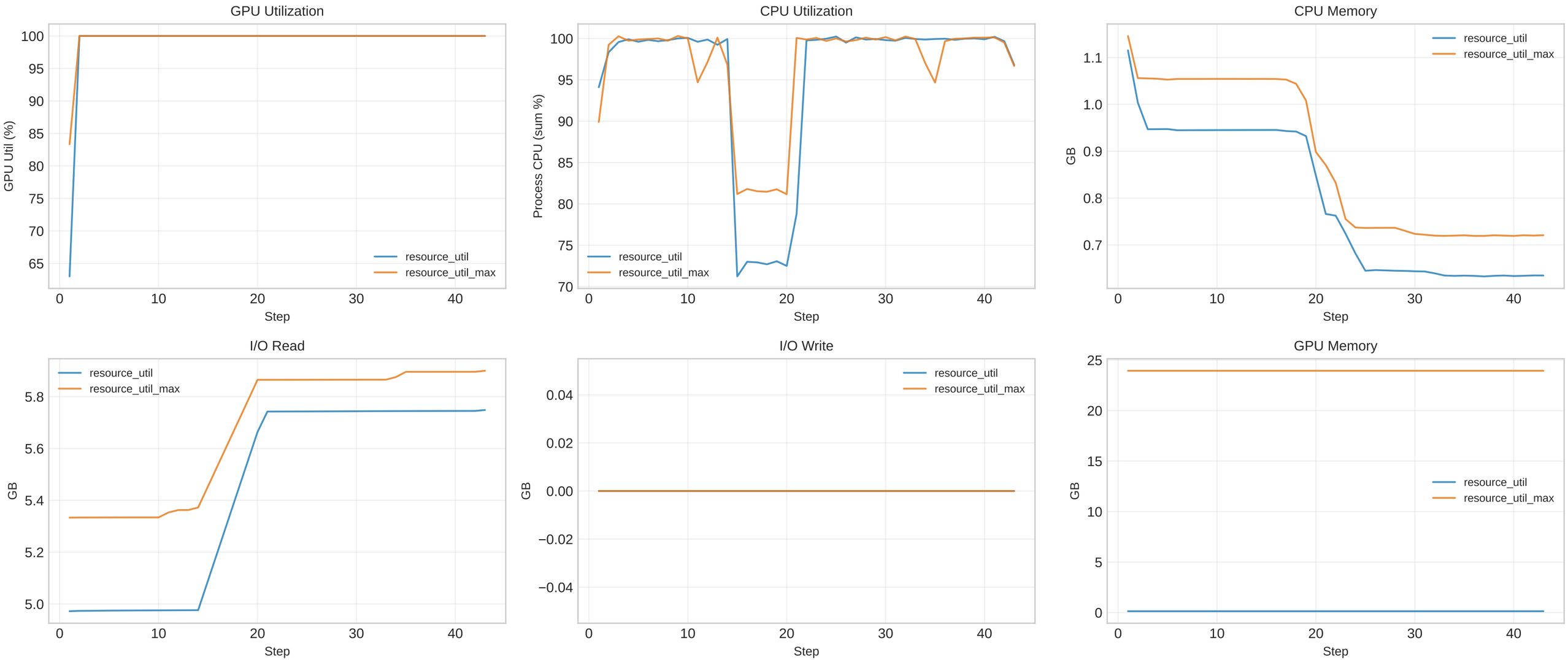
Metric Overlay — Disk, w=4, B=32
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



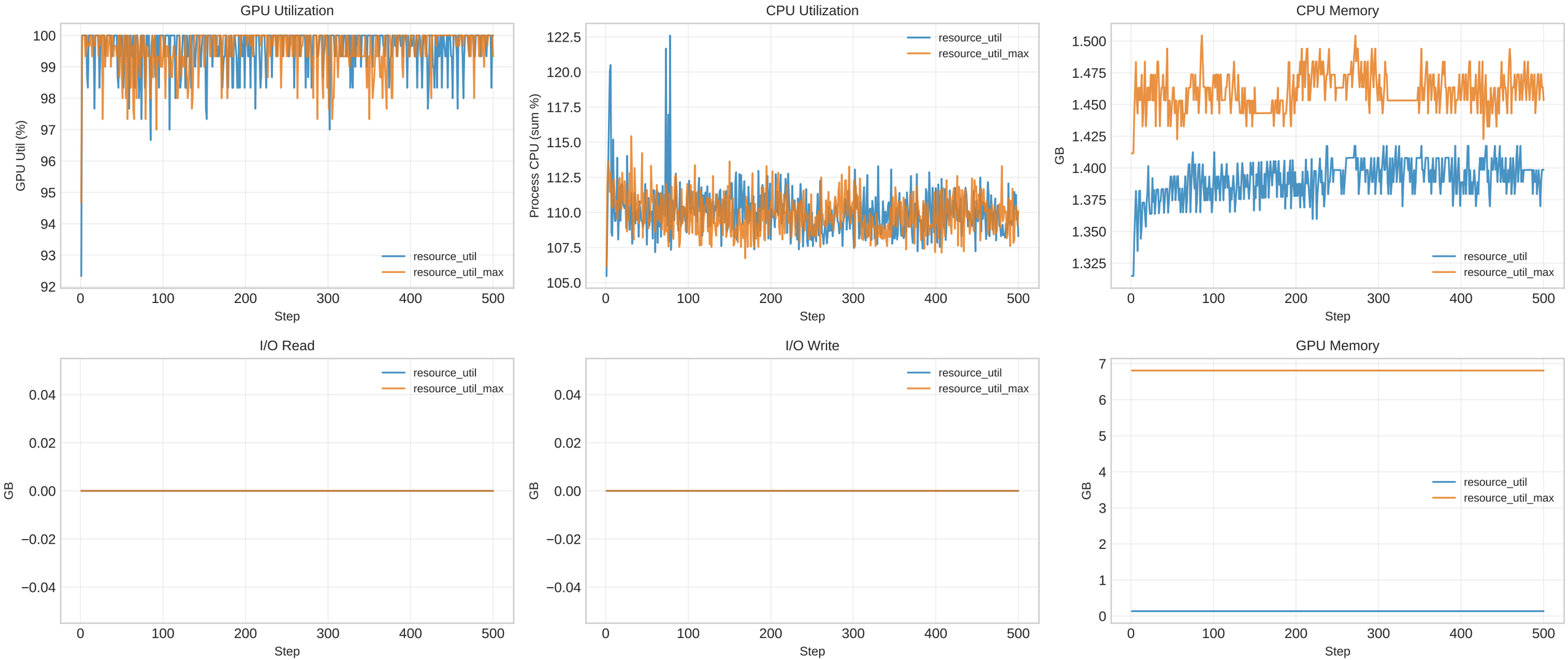
Metric Overlay — Disk, w=4, B=64
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



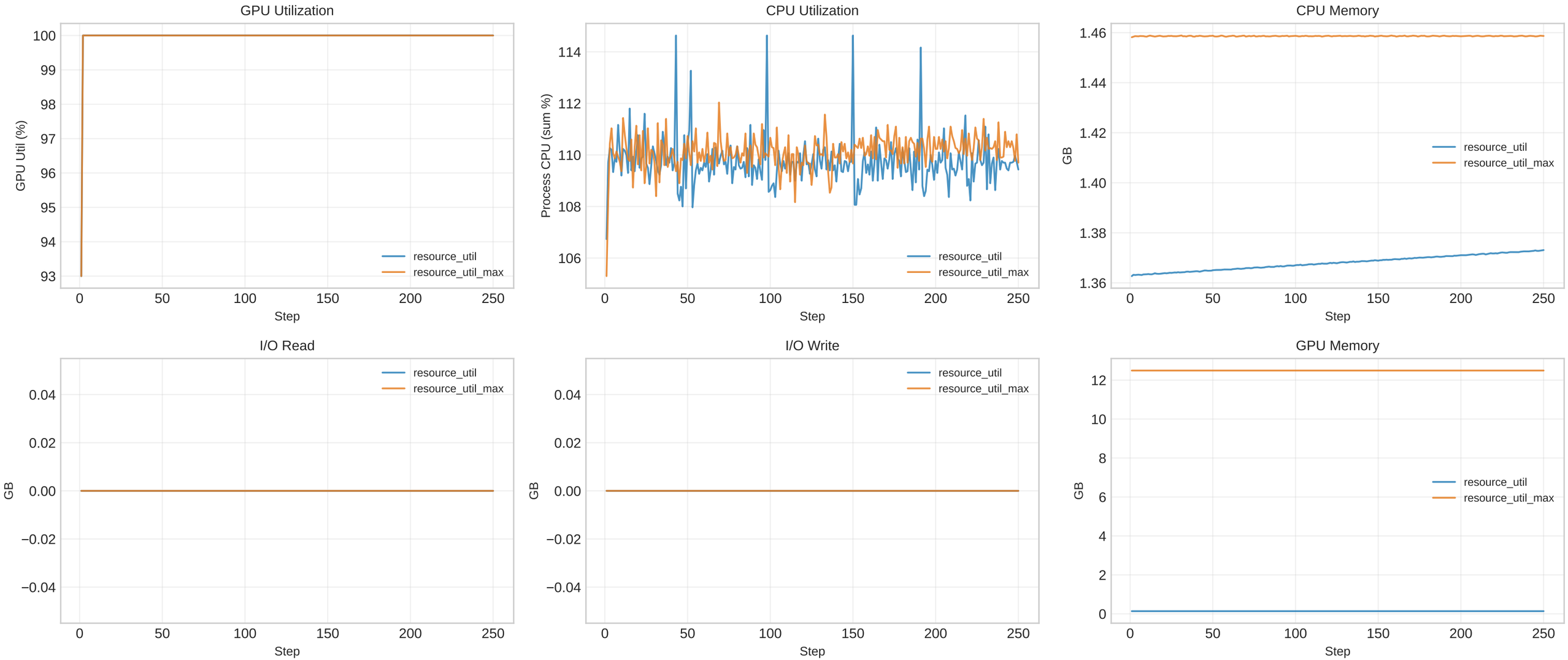
Metric Overlay — Disk, w=4, B=128
blue = resource_util (with CUDA sync) **orange** = resource_util_max (no sync)



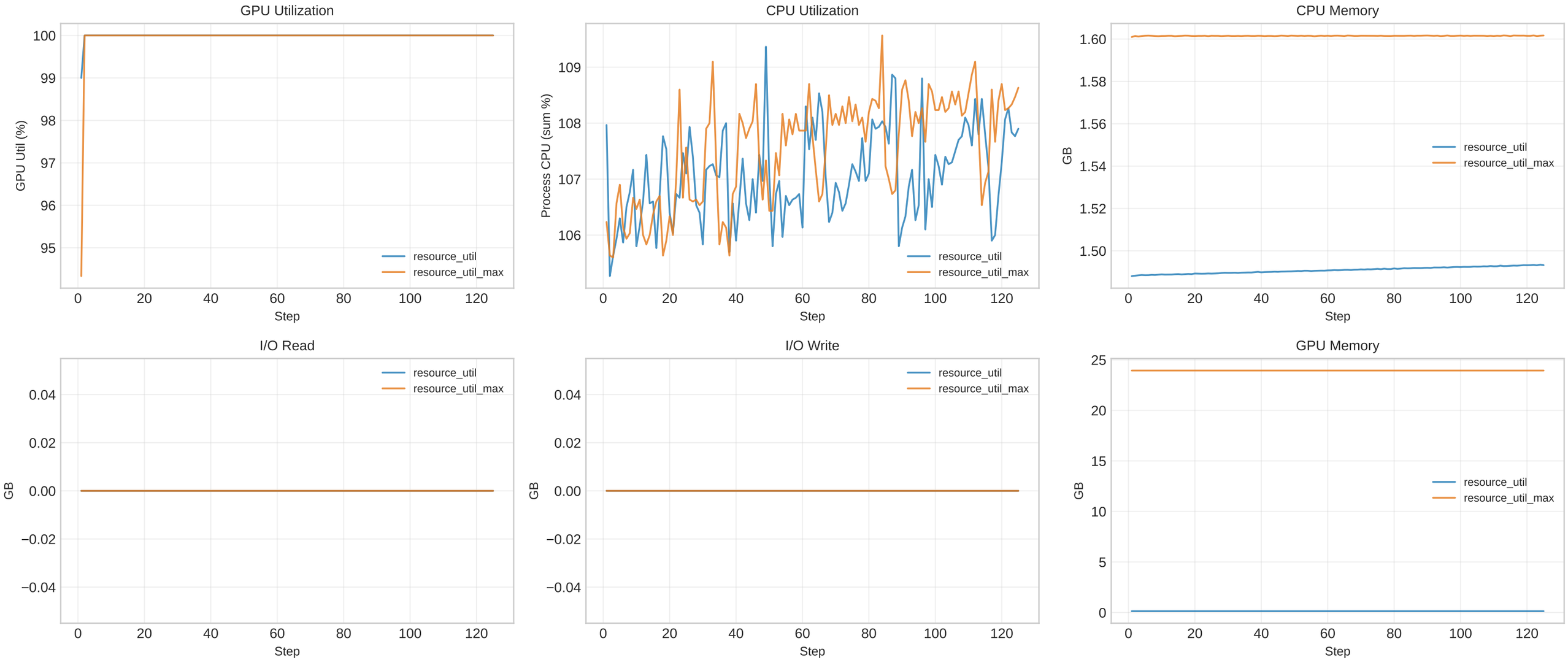
Metric Overlay — Milabench, w=0, B=32
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



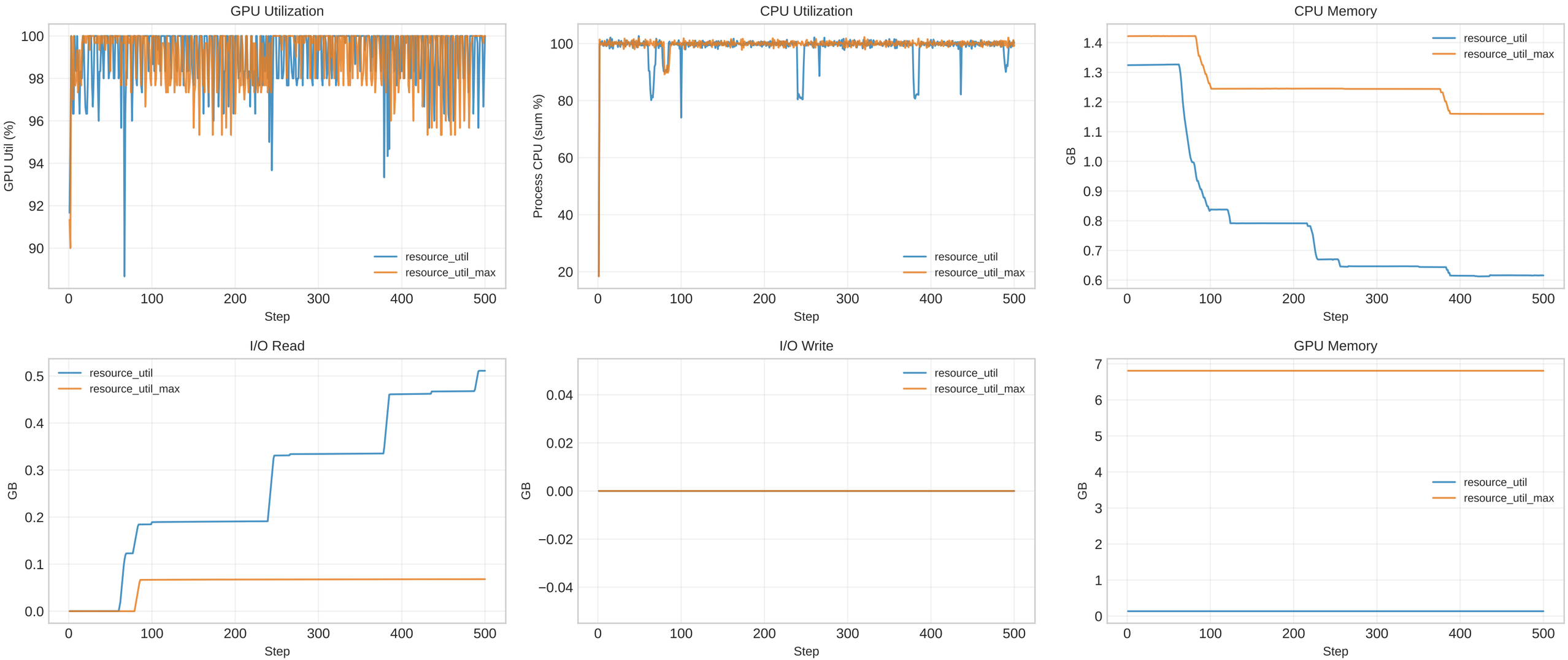
Metric Overlay — Milabench, w=0, B=64
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



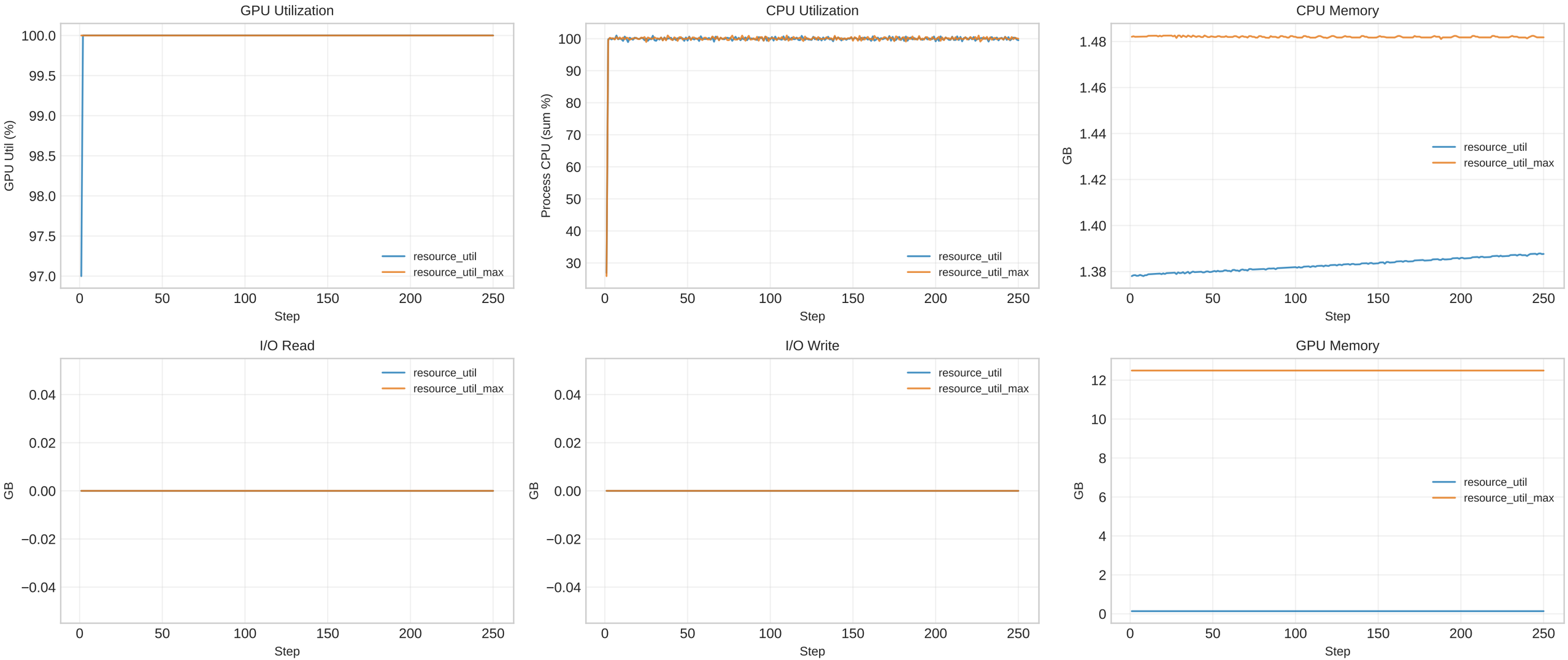
Metric Overlay — Milabench, w=0, B=128
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)



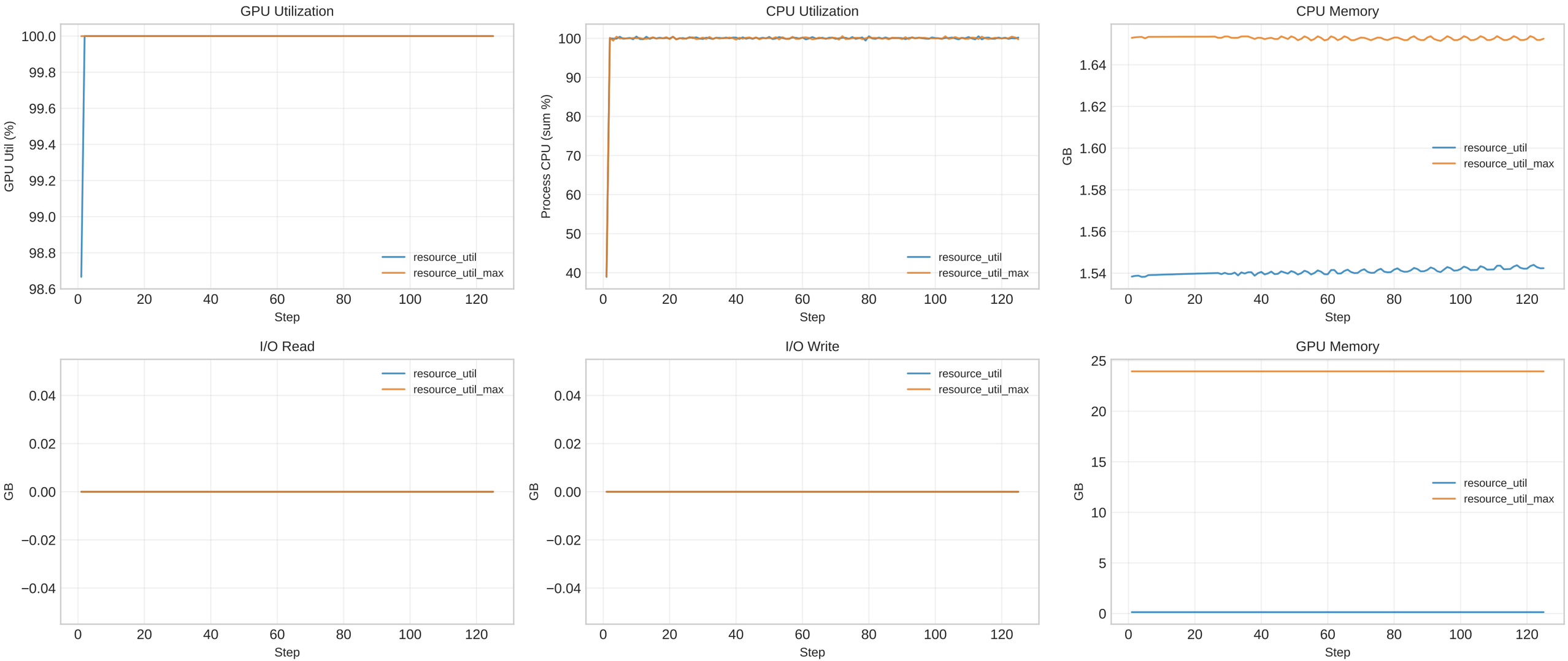
Metric Overlay — Milabench, w=4, B=32
blue = resource_util (with CUDA sync) **orange** = resource_util_max (no sync)



Metric Overlay — Milabench, w=4, B=64
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)

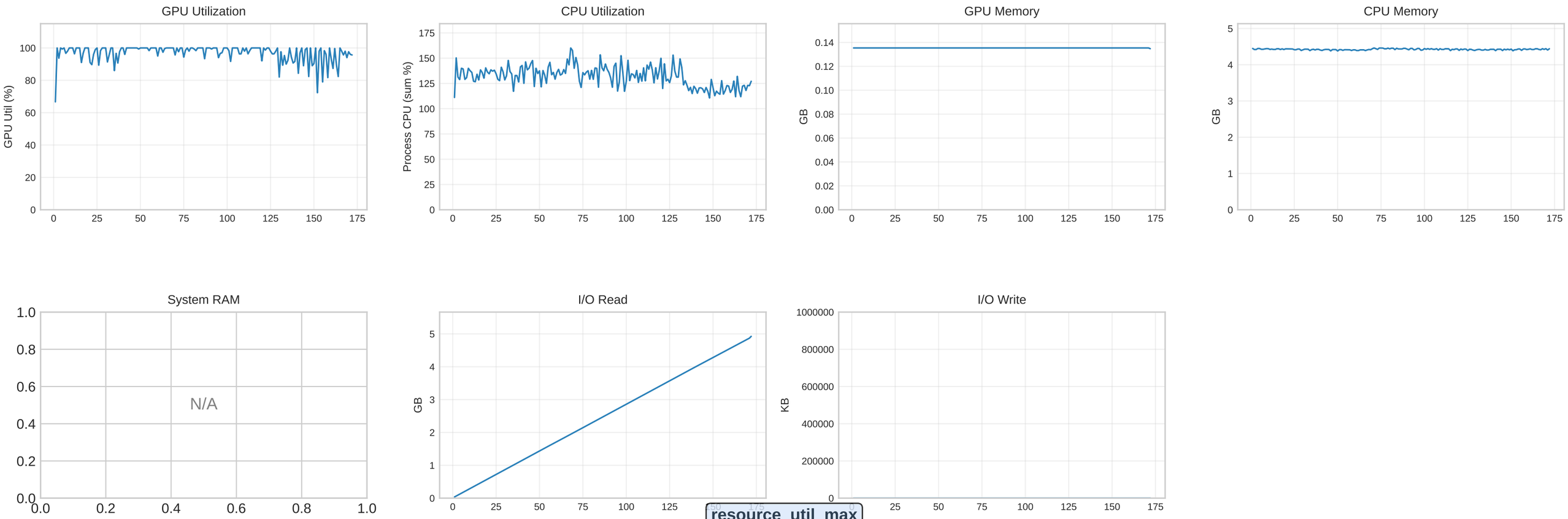


Metric Overlay — Milabench, w=4, B=128
blue = resource_util (with CUDA sync) orange = resource_util_max (no sync)

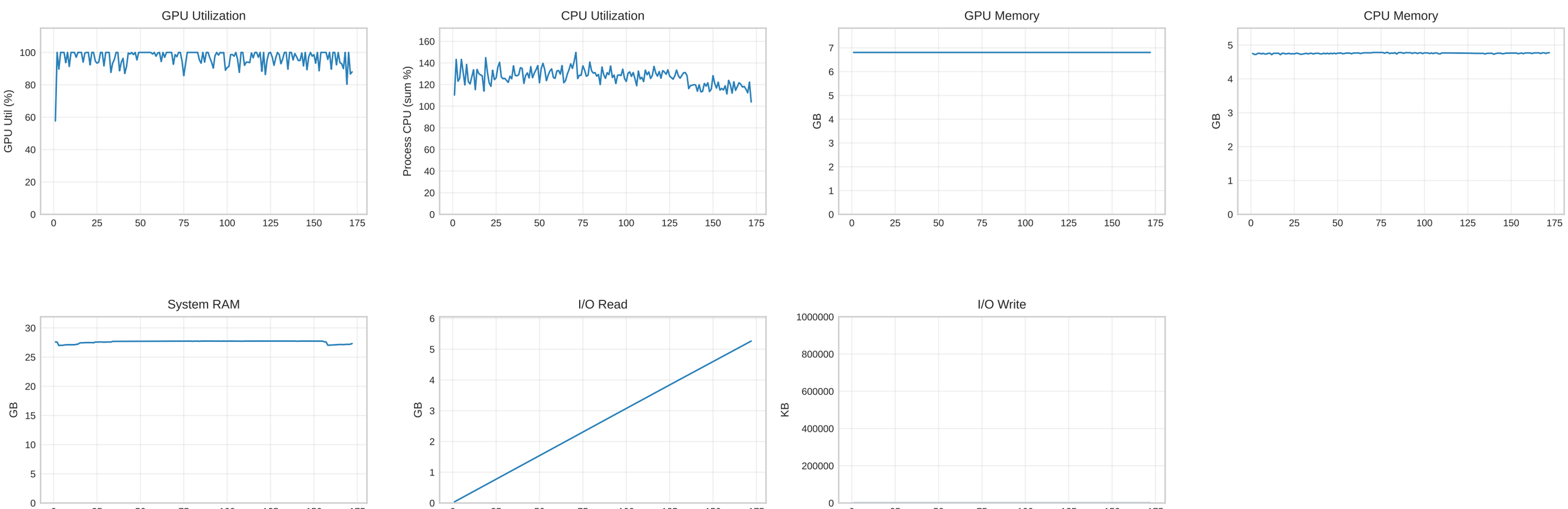


Resource Timeline — Disk, w=0, B=32

resource_util

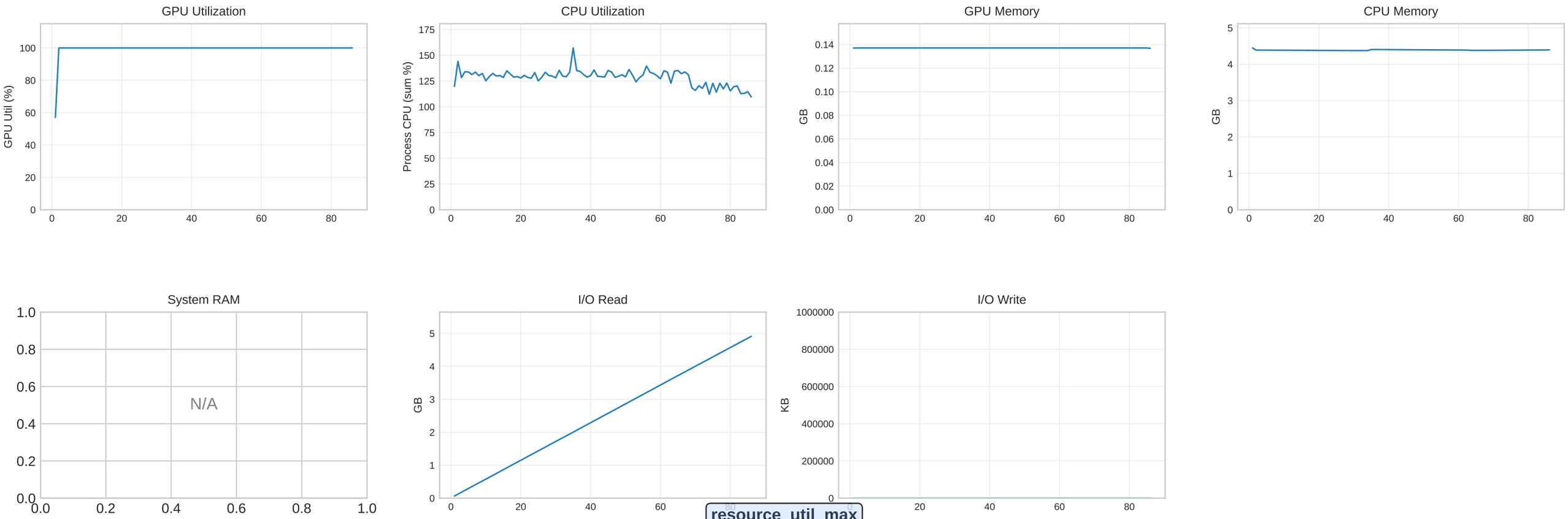


resource_util_max

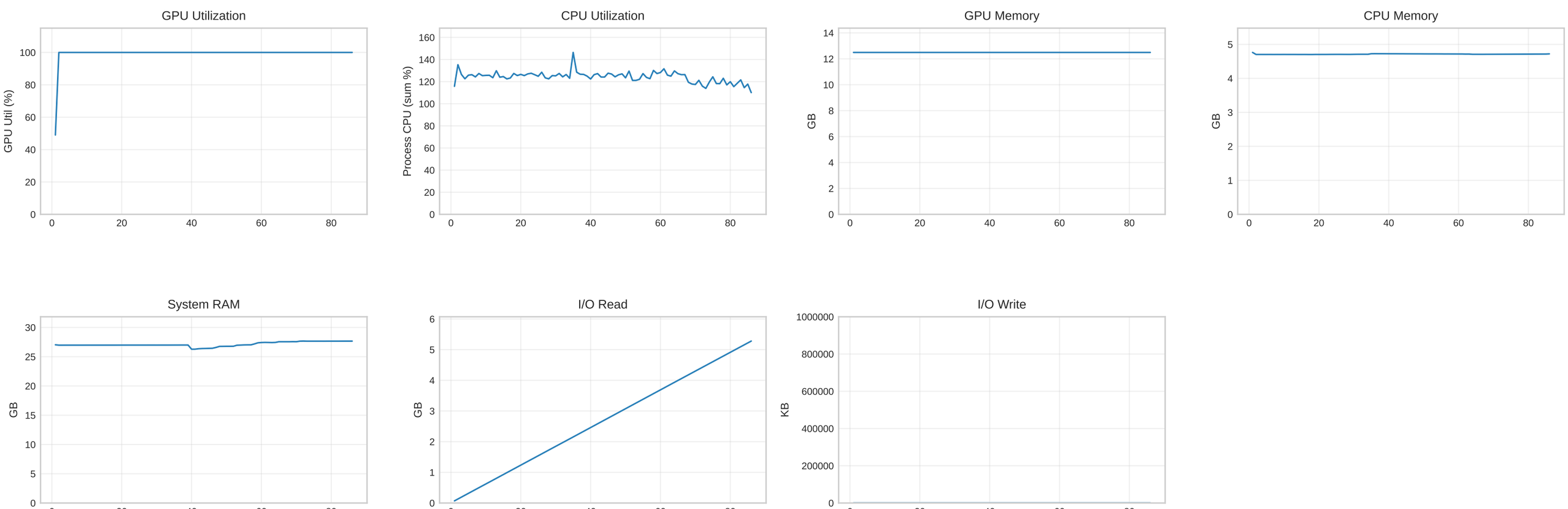


Resource Timeline — Disk, w=0, B=64

resource_util

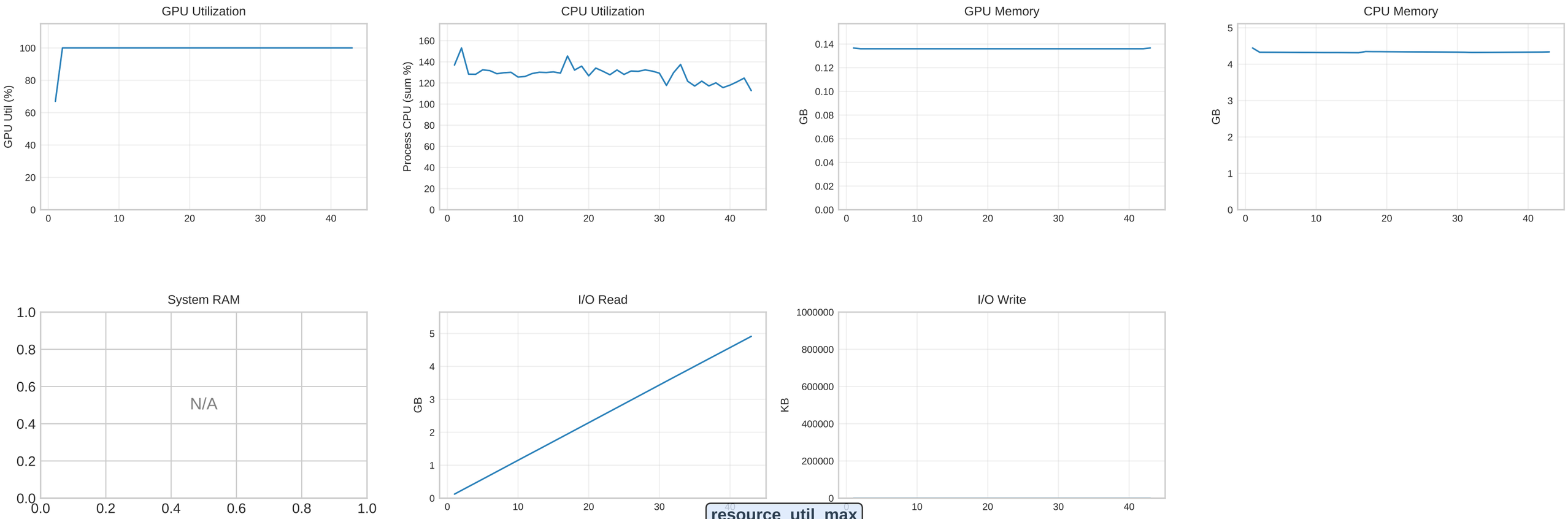


resource_util_max

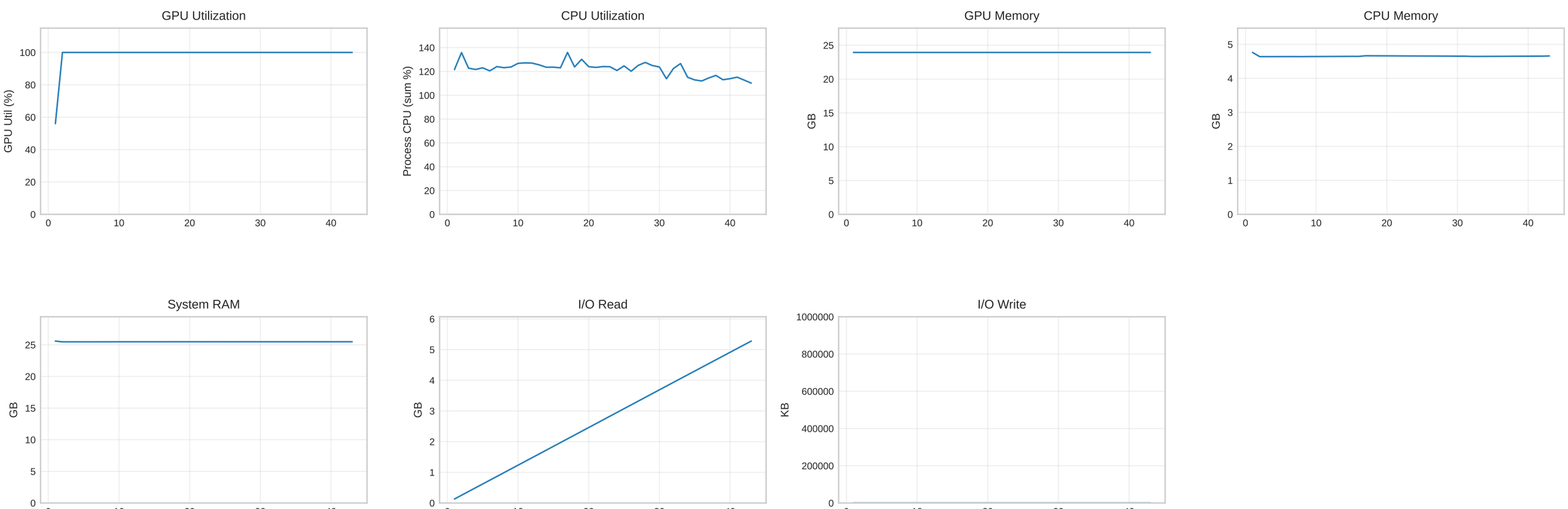


Resource Timeline — Disk, w=0, B=128

resource_util

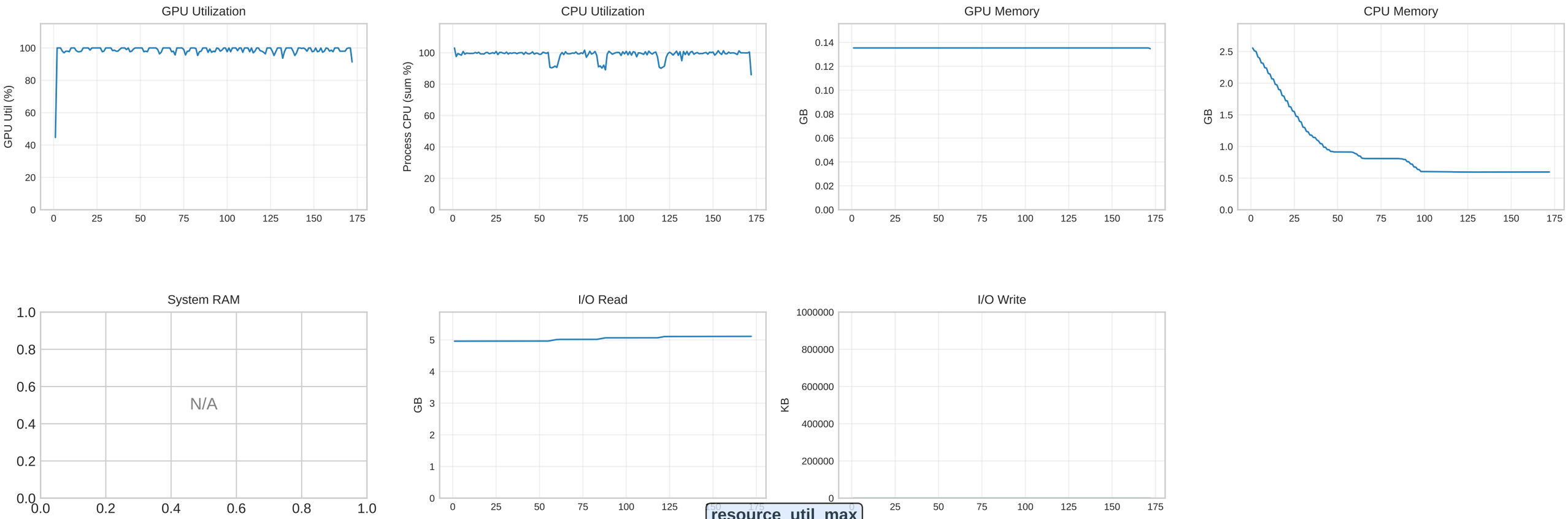


resource_util_max

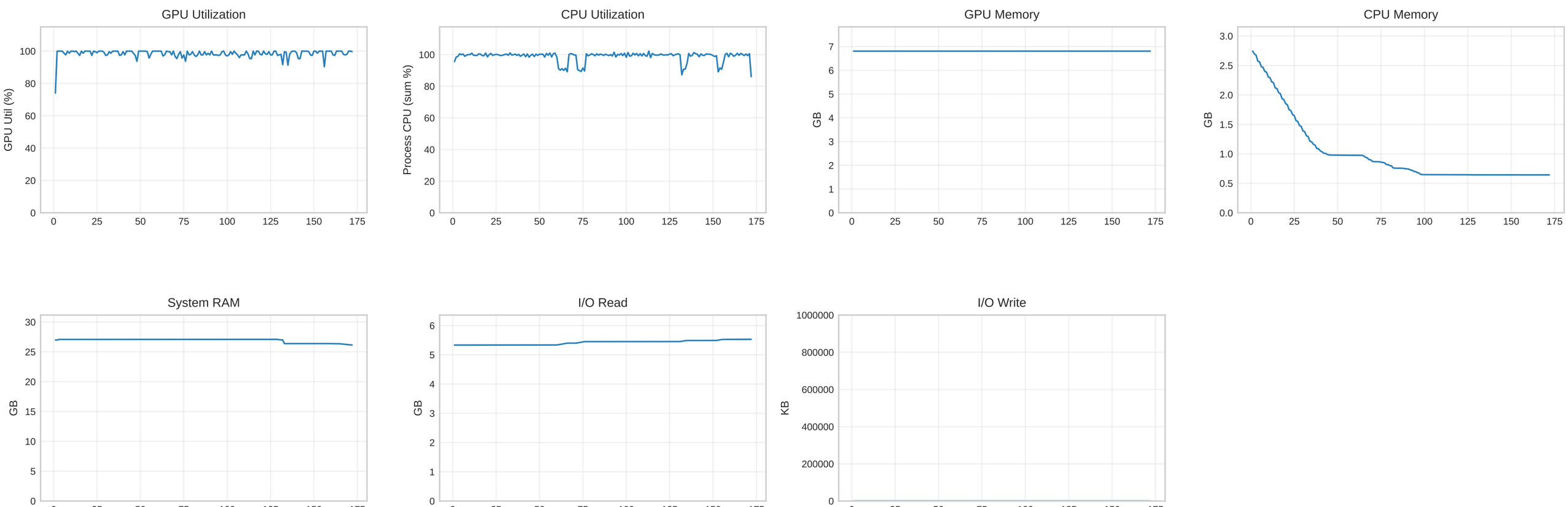


Resource Timeline — Disk, w=4, B=32

resource_util

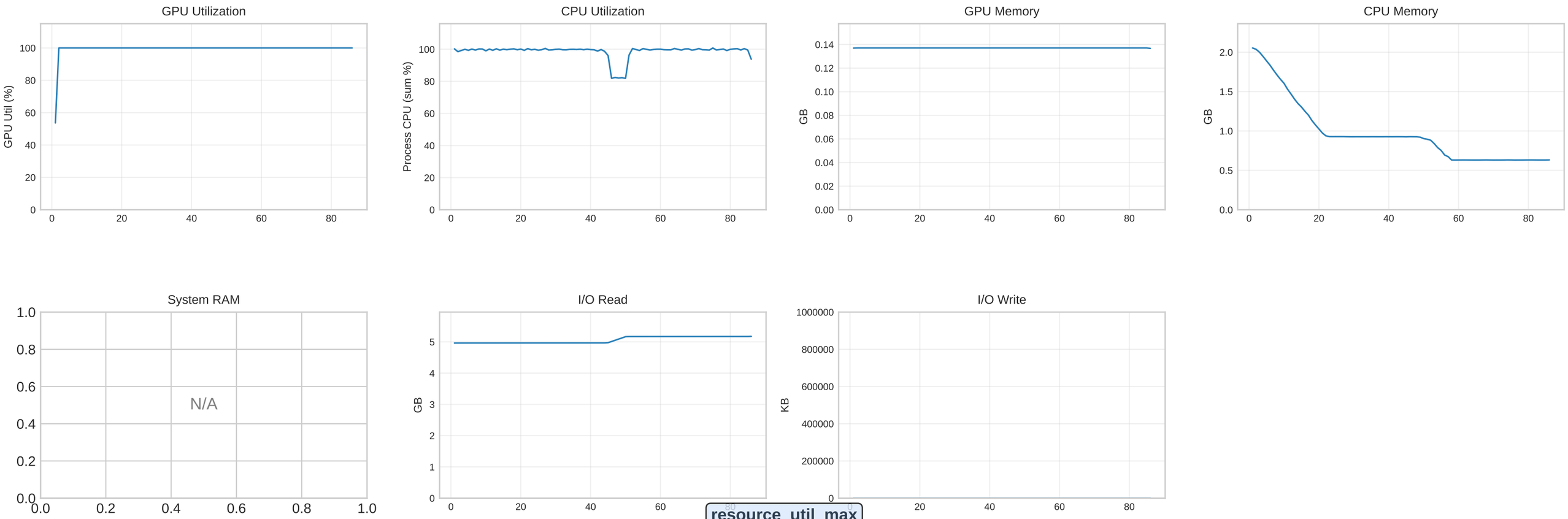


resource_util_max

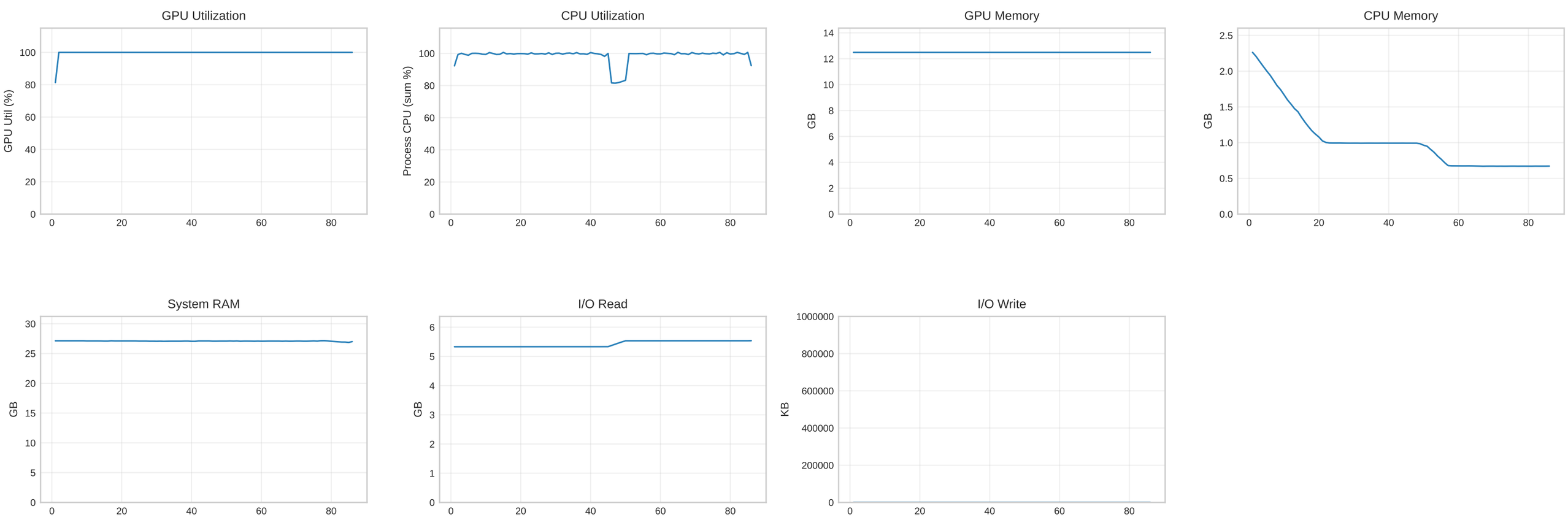


Resource Timeline — Disk, w=4, B=64

resource_util

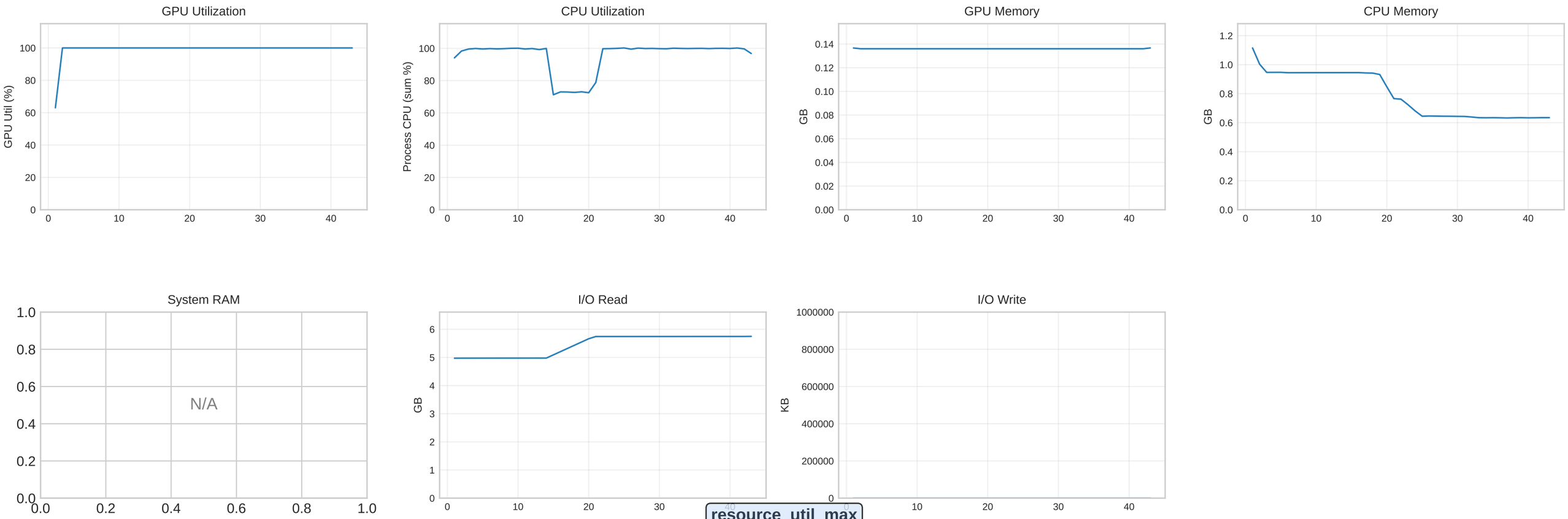


resource_util_max

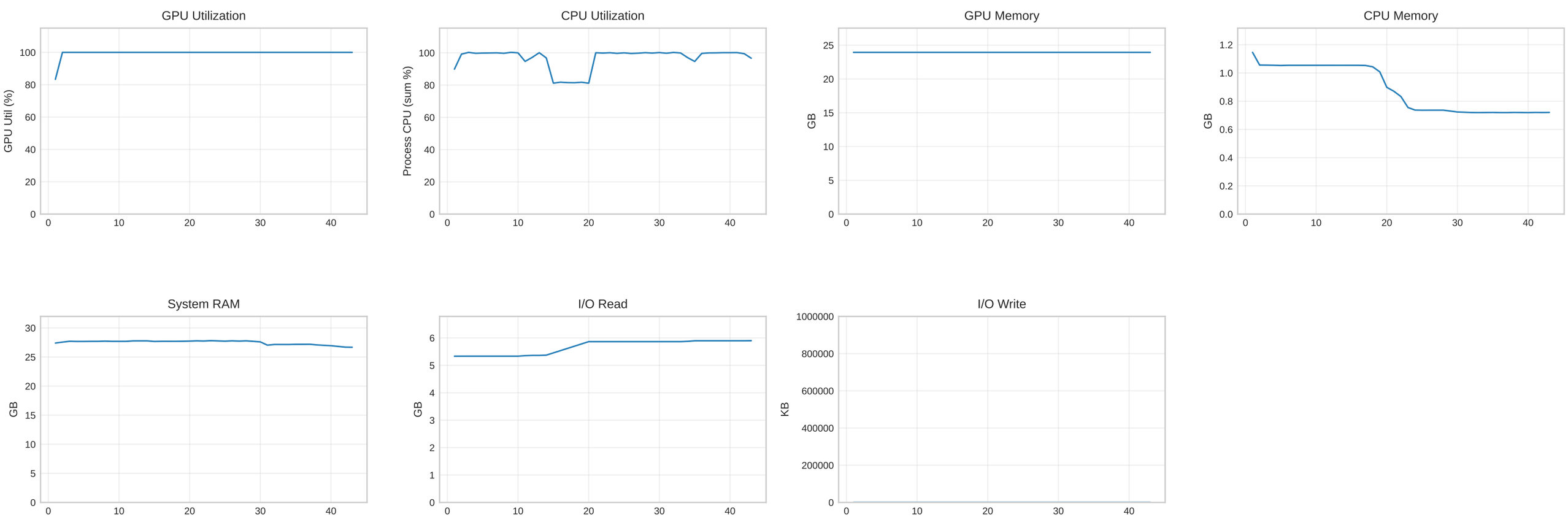


Resource Timeline — Disk, w=4, B=128

resource_util

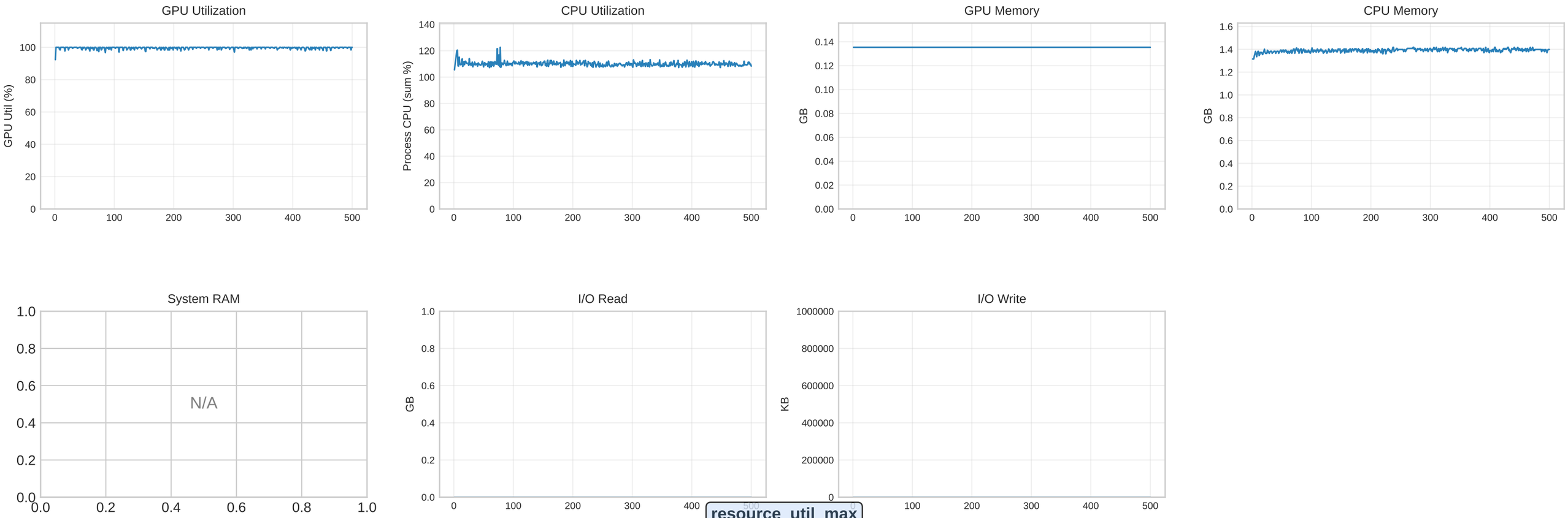


resource_util_max

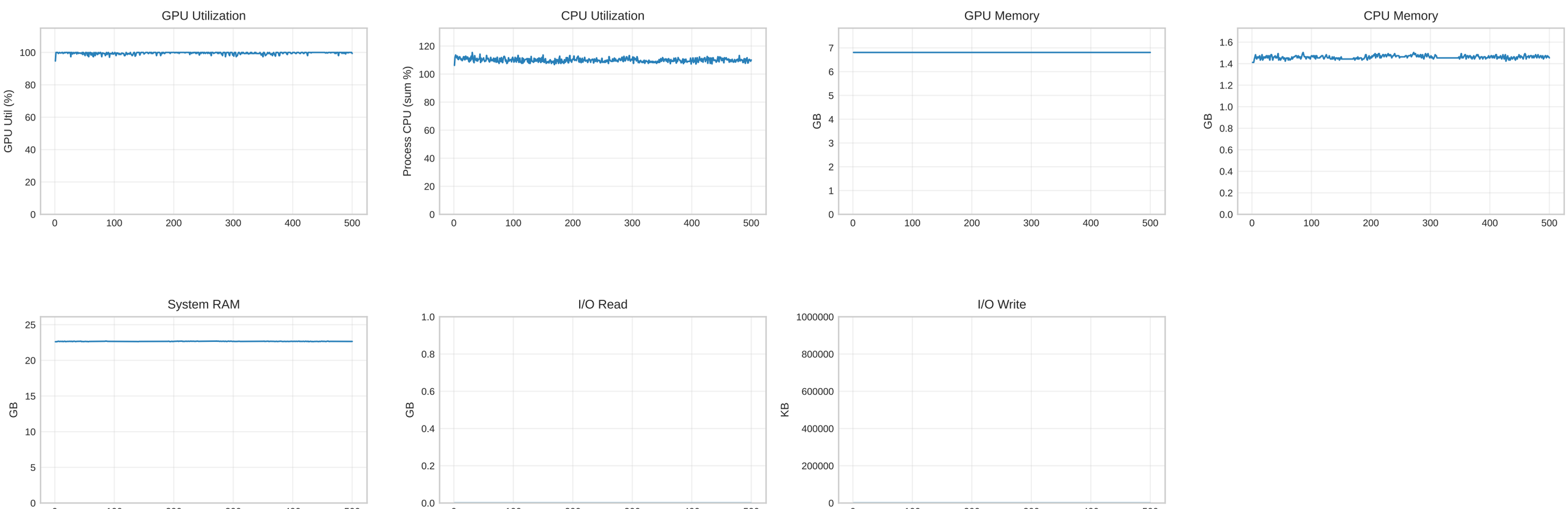


Resource Timeline — Milabench, w=0, B=32

resource_util

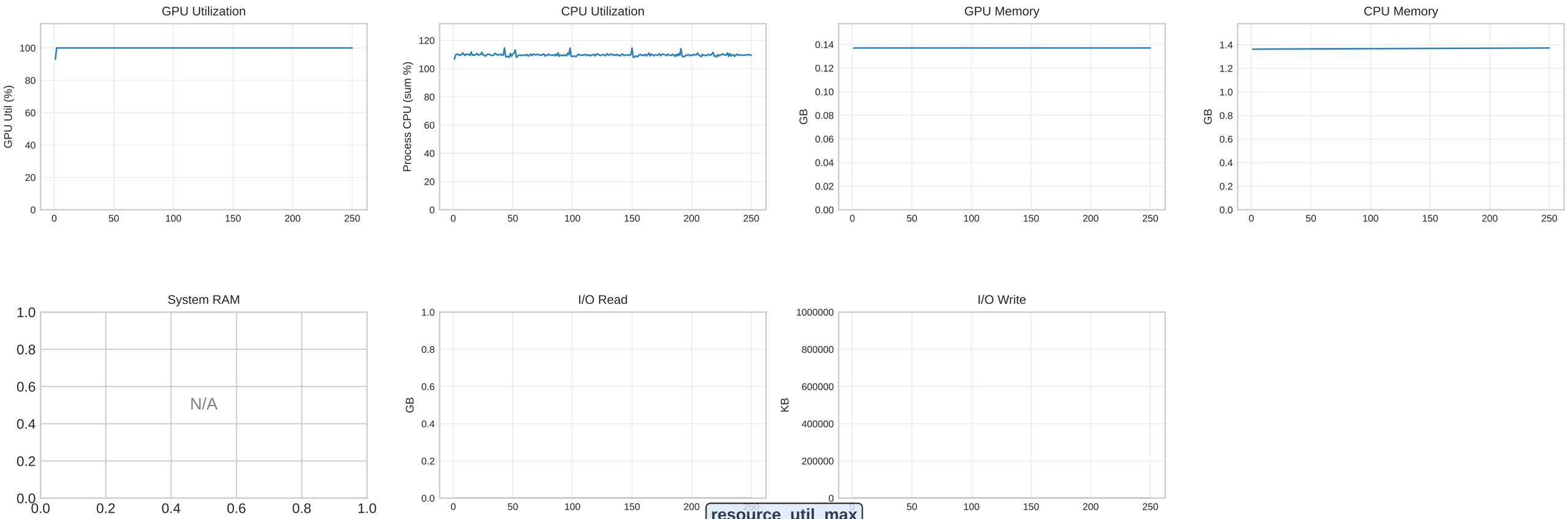


resource_util_max

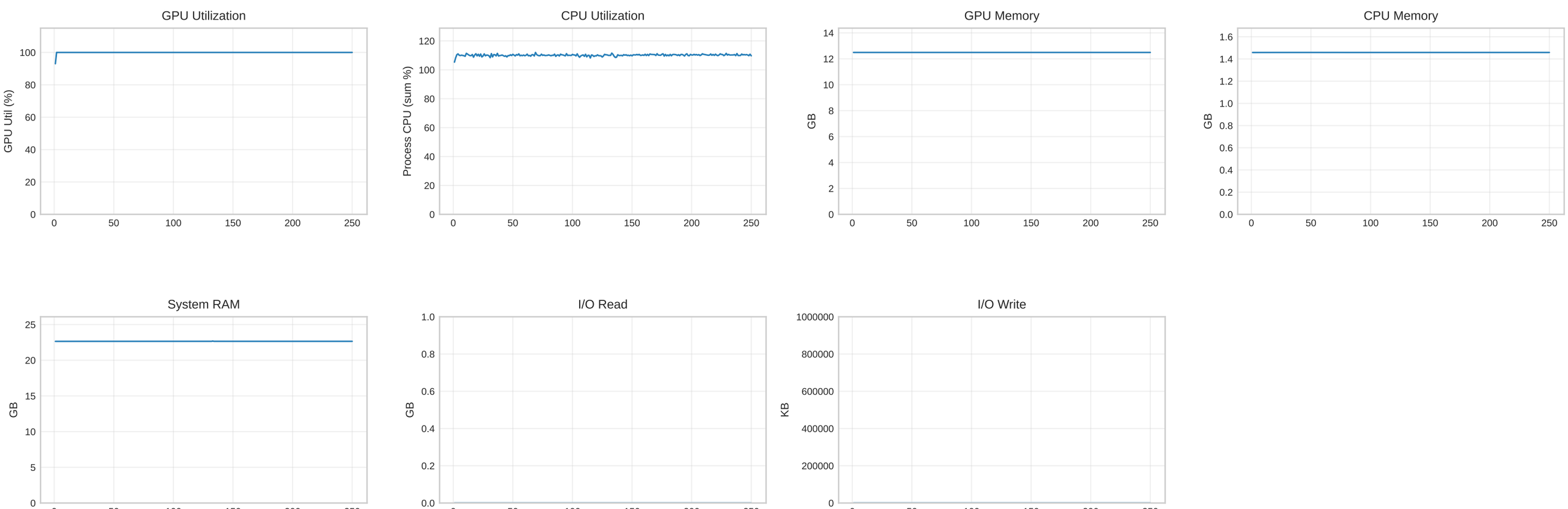


Resource Timeline — Milabench, w=0, B=64

resource_util

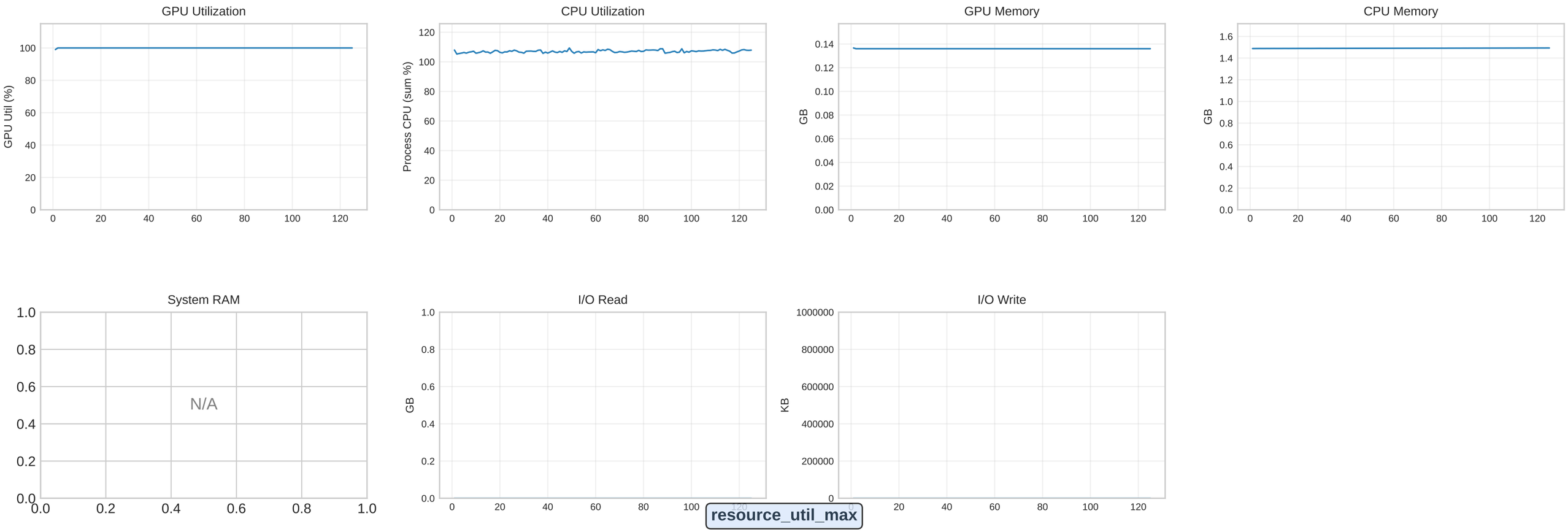


resource_util_max

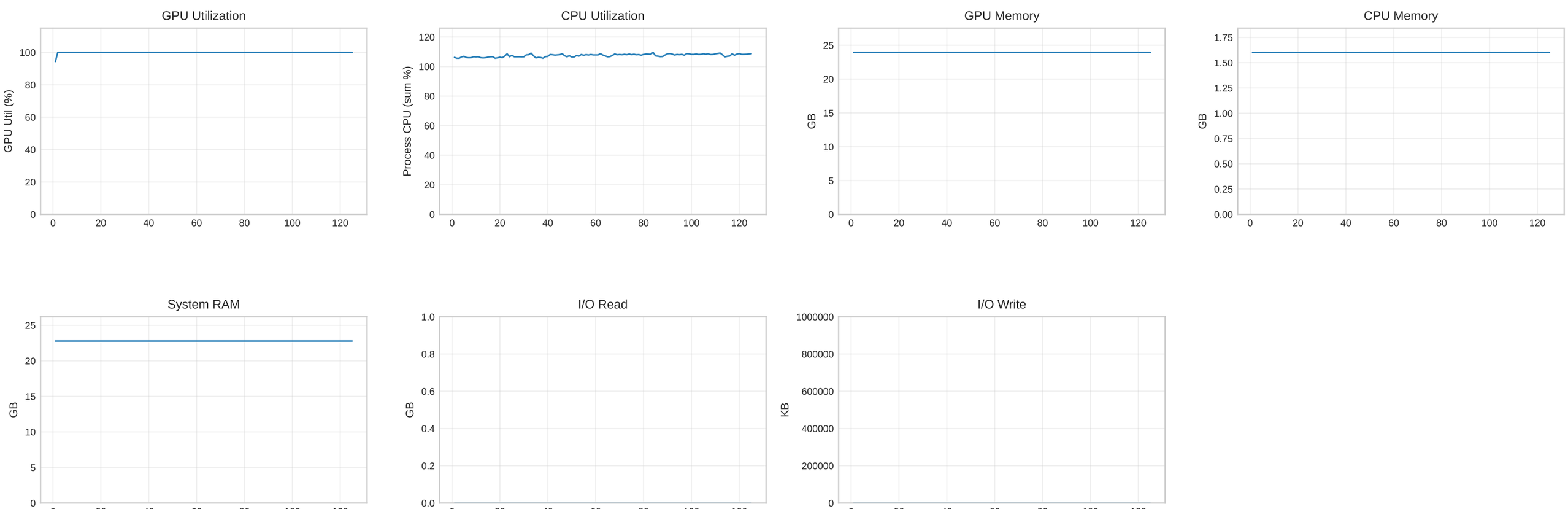


Resource Timeline — Milabench, w=0, B=128

resource_util

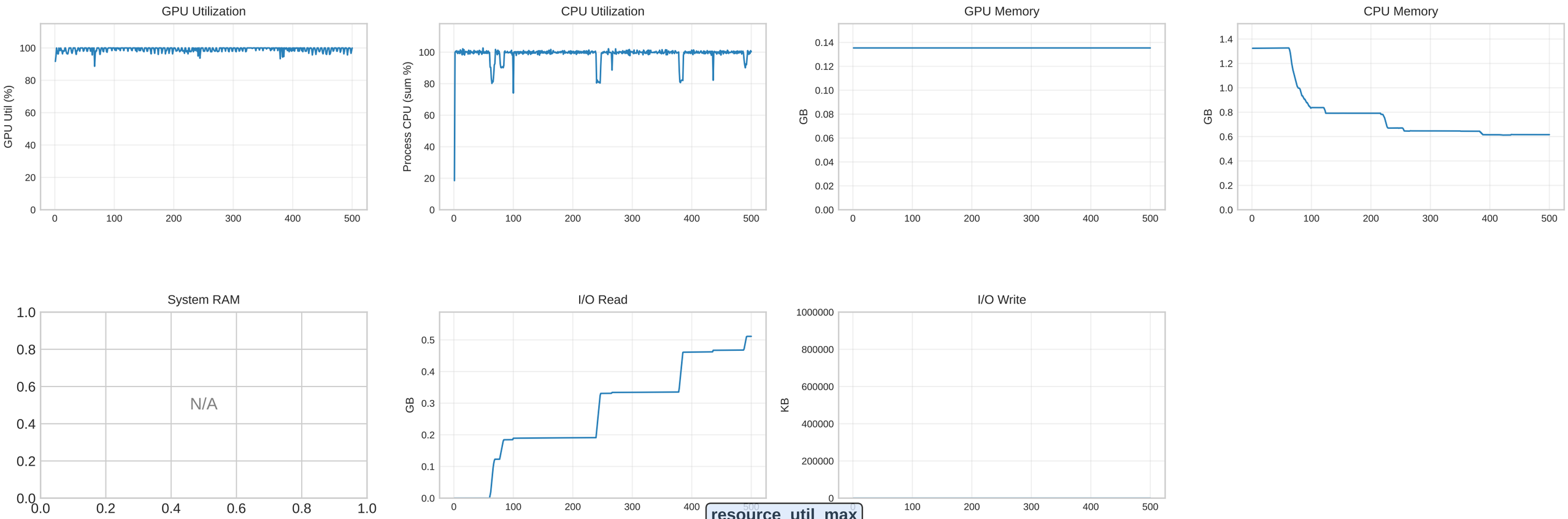


resource_util_max

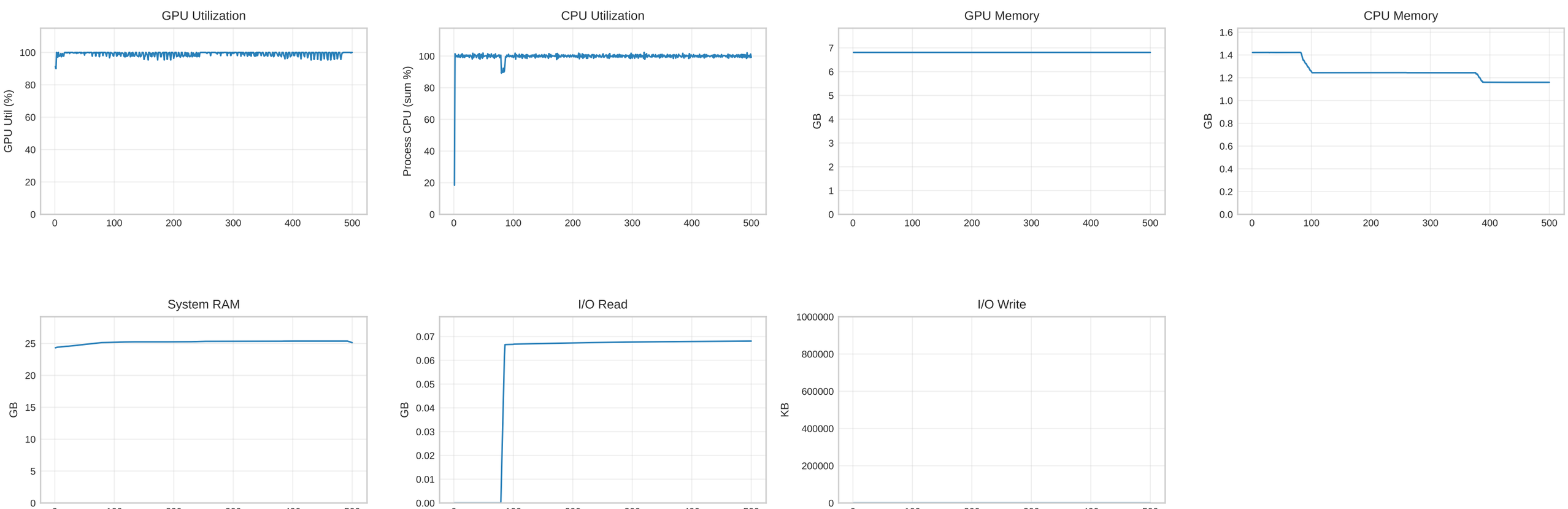


Resource Timeline — Milabench, w=4, B=32

resource_util

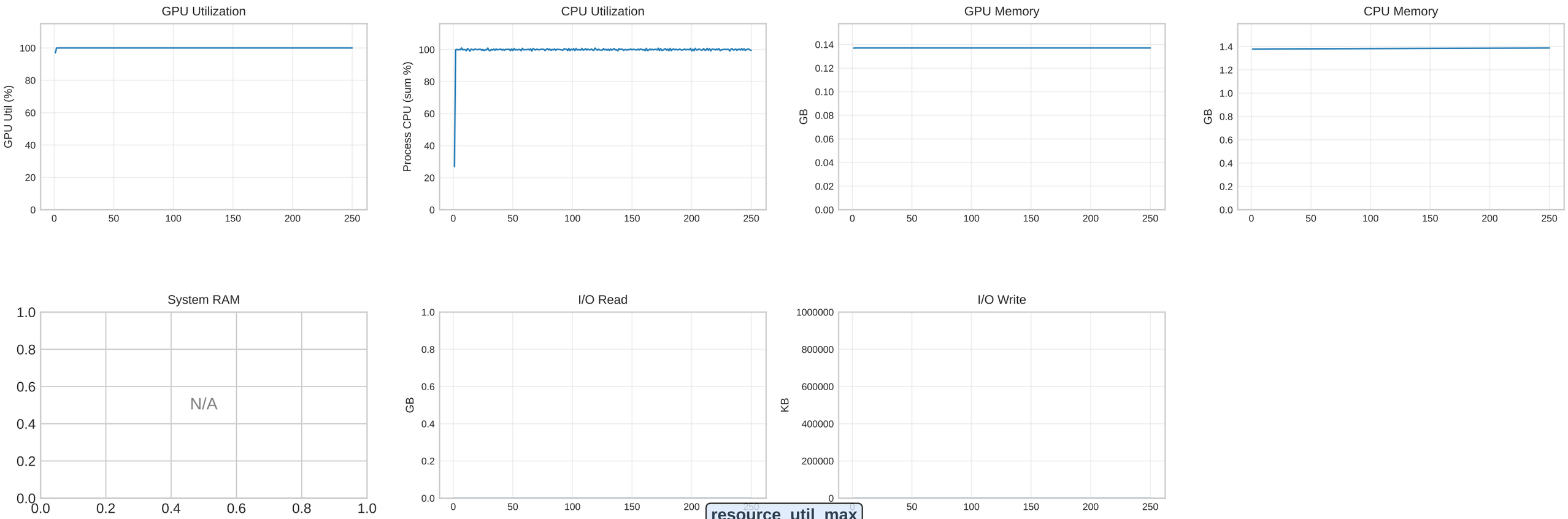


resource_util_max

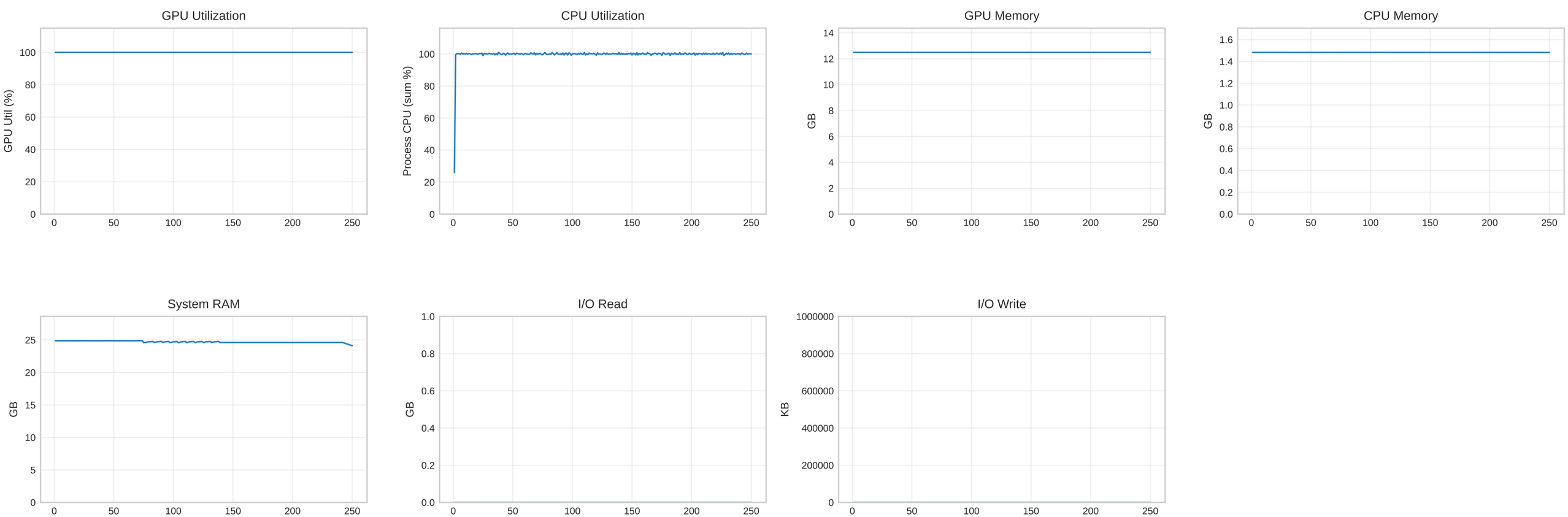


Resource Timeline — Milabench, w=4, B=64

resource_util

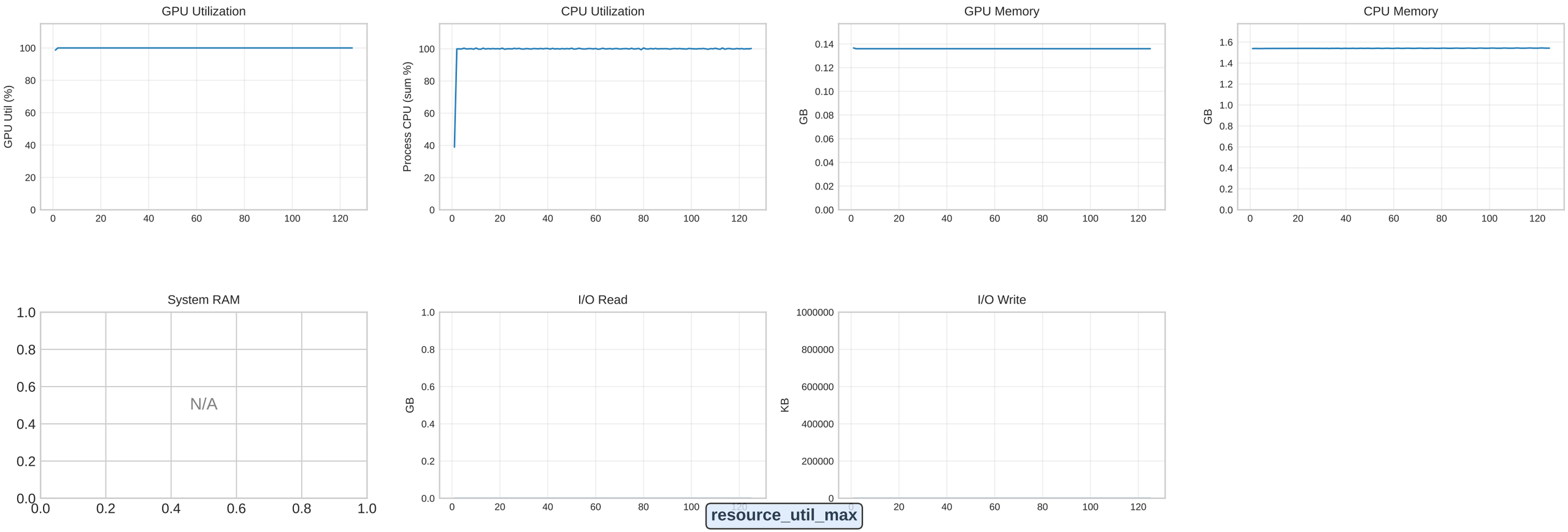
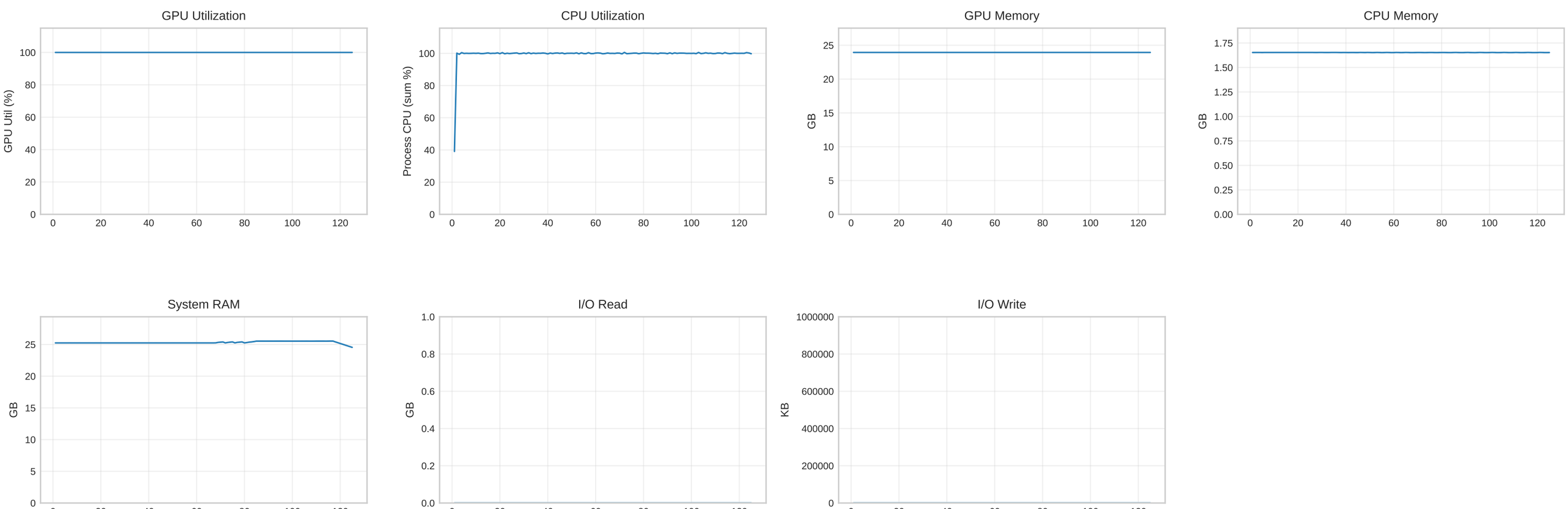


resource_util_max



Resource Timeline — Milabench, w=4, B=128

resource_util



resource_util_max